MOTION SUIT

MS2 18/11/2025

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Projeto em Engenharia Informática 2025/2026

MOTION SUIT

Our project involves developing a suit equipped with sensors that measure body posture and detect anomalies

RELATED WORK

	SmartPosture	FlexiTrace	SwordHealth	DorsaVI	Current Suit	MotionSuit
Performance Analytics	✓	✓	✓	✓	✓	✓
Real-Time Feedback	✗	✗	✓	✓	✓	✓
Wearable Sensors	✗	✗	✗	✓	✓	✓
Gamification	✗	✓	✗	✗	✗	✓

RELATED WORK

	SmartPosture	FlexiTrace	SwordHealth	DorsaVI
Language				
Frameworks				.NET Framework
Cloud & Infrastructure		aws		aws
APIs & Integrations	—	—		—

USER STORIES

	As a	I want	So that
	Suit User	To receive alerts about my posture	I can prevent injuries in the future
	Researcher	To analyze the captured data	I can identify postural deviations and abnormal patterns
	Administrator	To manage user accounts	I can control acess and provide support

FUNCTIONAL

SUIT USER

01 The system must allow to register and log in with their respective role

02 The system must deliver posture-related alerts to users

03 The system must provide a dashboard displaying all personal user data

04 The system must enable users to export their personal data

FUNCTIONAL

RESEARCHER

05 The system must display all user data with multiple views and selectable time periods.

06 The system must allow researchers to export data from all users.

FUNCTIONAL

ADMINISTRATOR

07 The system must provide user account management capabilities

08 The system must provide an administrative dashboard displaying system metrics.

NON-FUNCTIONAL

Security and Privacy

01 The system must encrypt all the Protected Health Information

02 The system must store passwords using salted cryptographic hashes

Reliability

Usability and User Experience

Data Management

Maintainability and Scalability

NON-FUNCTIONAL

Security and Privacy

Reliability

03 The system must guarantee zero data loss at the end of each session

04 The system must support offline operability through local data caching

Usability and User Experience

Data Management

Maintainability and Scalability

NON-FUNCTIONAL

Security and Privacy

Reliability

Usability and User Experience

05 The system must allow new users to perform basic monitoring tasks

06 The system must enable users to personalize dashboard layouts

Data Management

Maintainability and Scalability

NON-FUNCTIONAL

Security and Privacy

Reliability

Usability and User Experience

Data Management

07 The system must ensure high-performance queries for user-facing operations

08 The system must provide reliable retention of raw sensor data

Maintainability and Scalability

NON-FUNCTIONAL

Security and Privacy

Reliability

Usability and User Experience

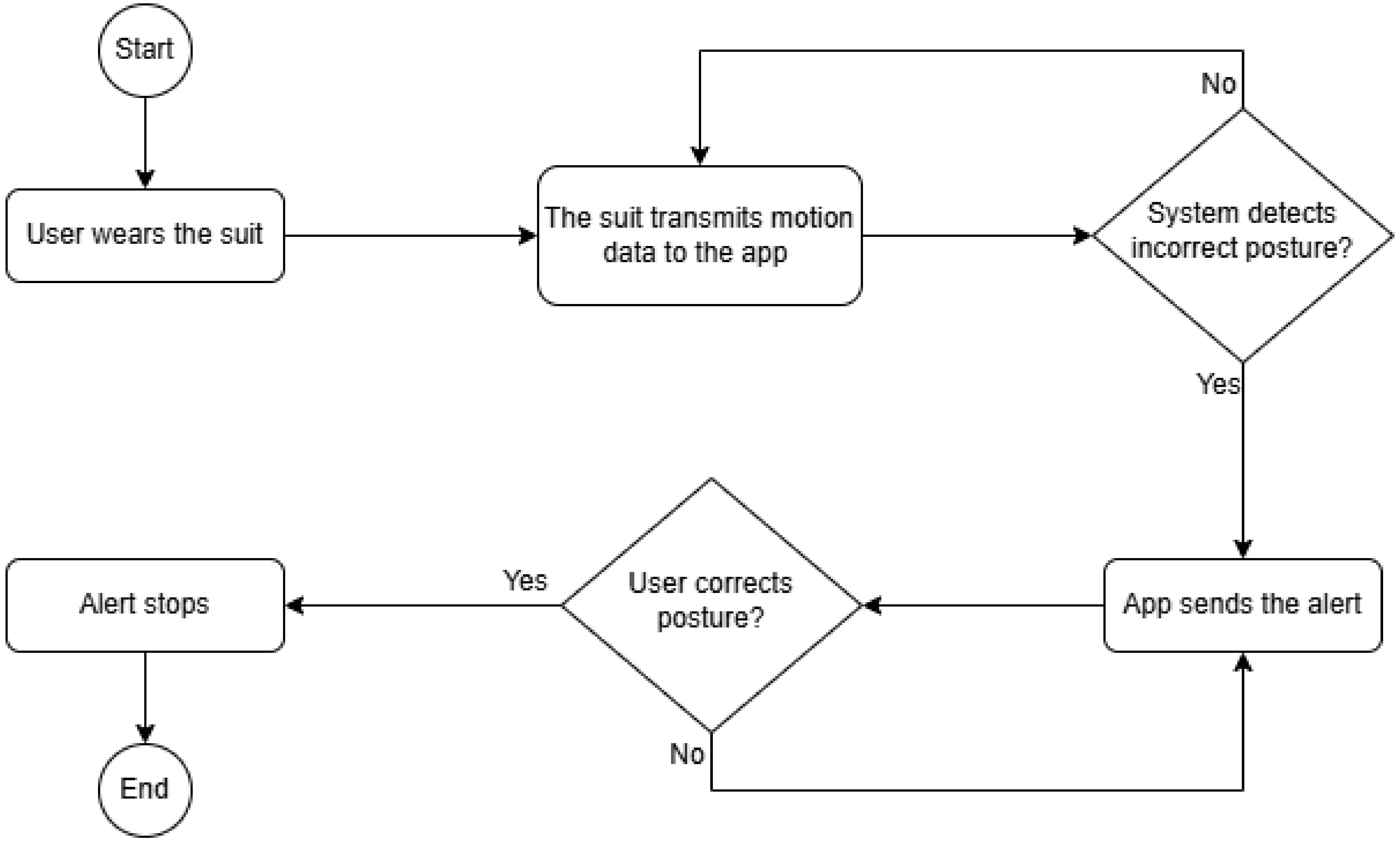
Data Management

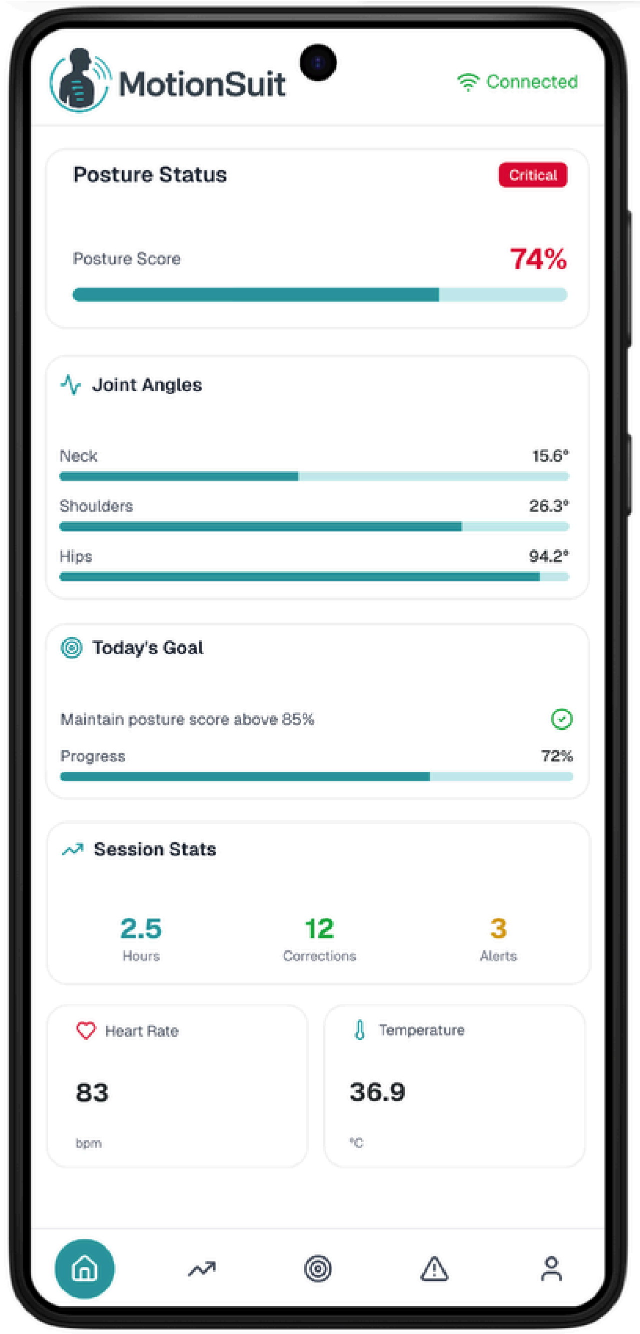
Maintainability and Scalability

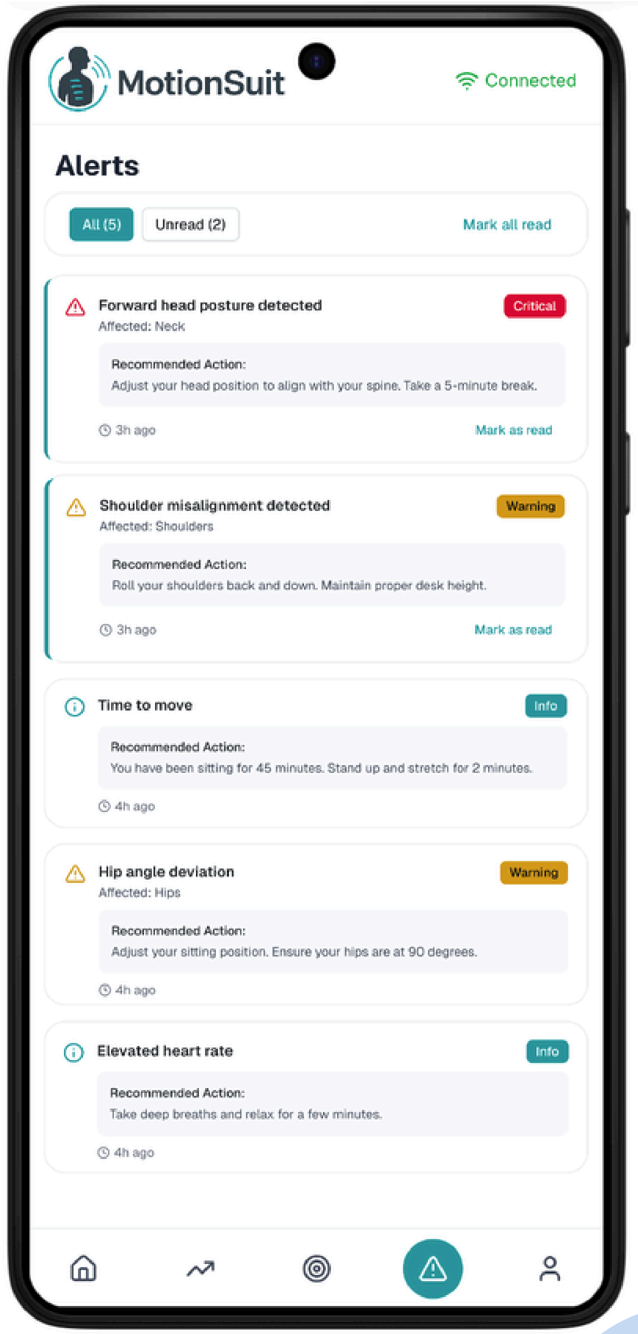
09 The system must expose well-defined APIs for all components

10 The system must support the addition of new sensor types through its sensor integration

UC-1 → Receive Posture Alerts







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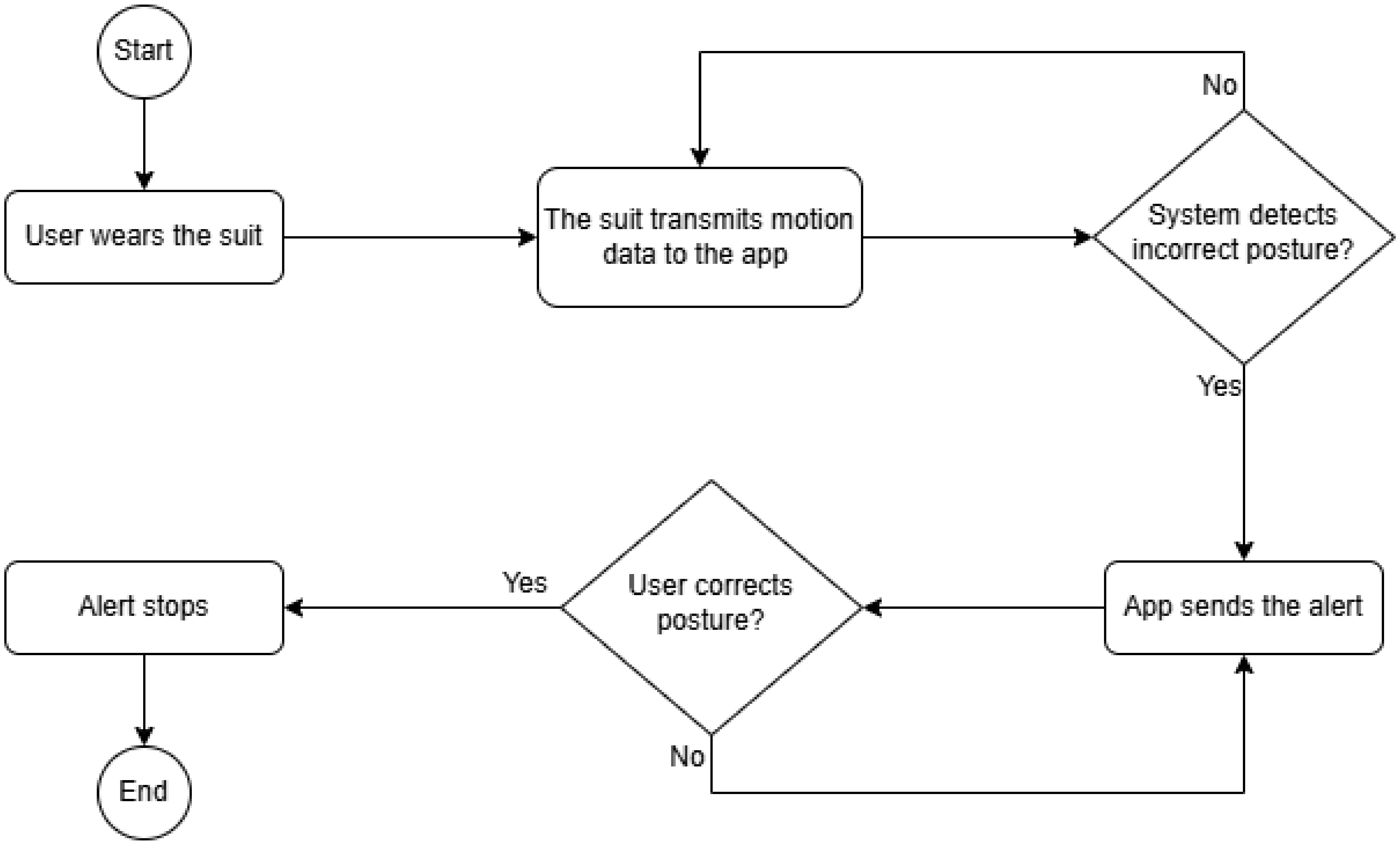

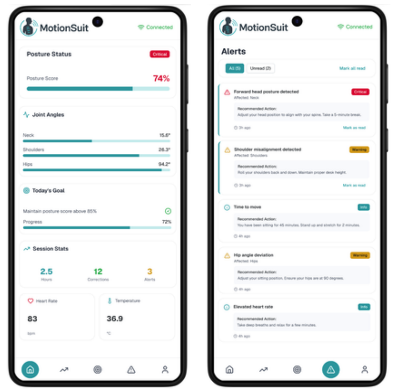

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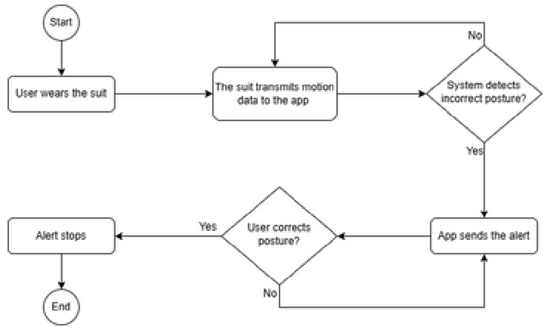
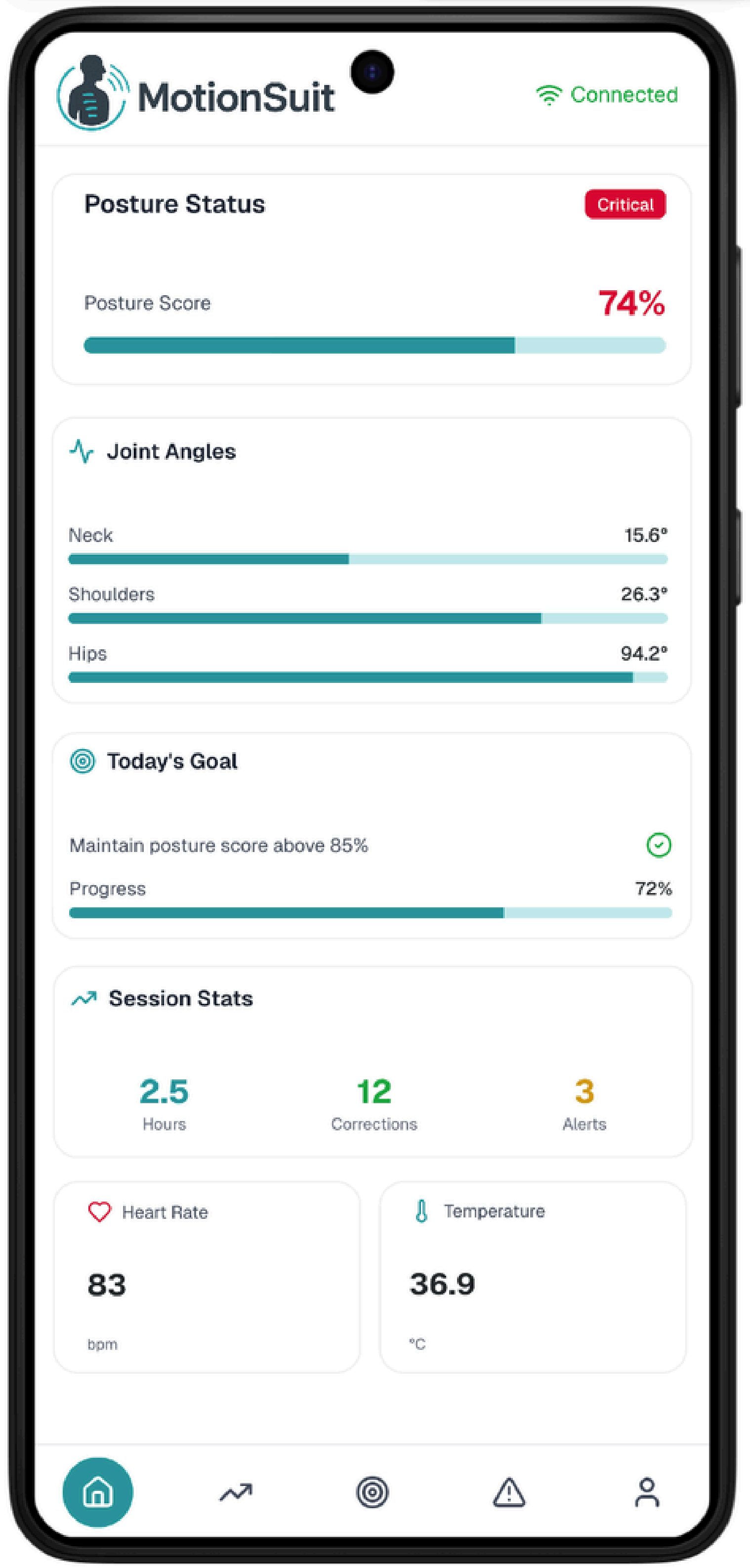
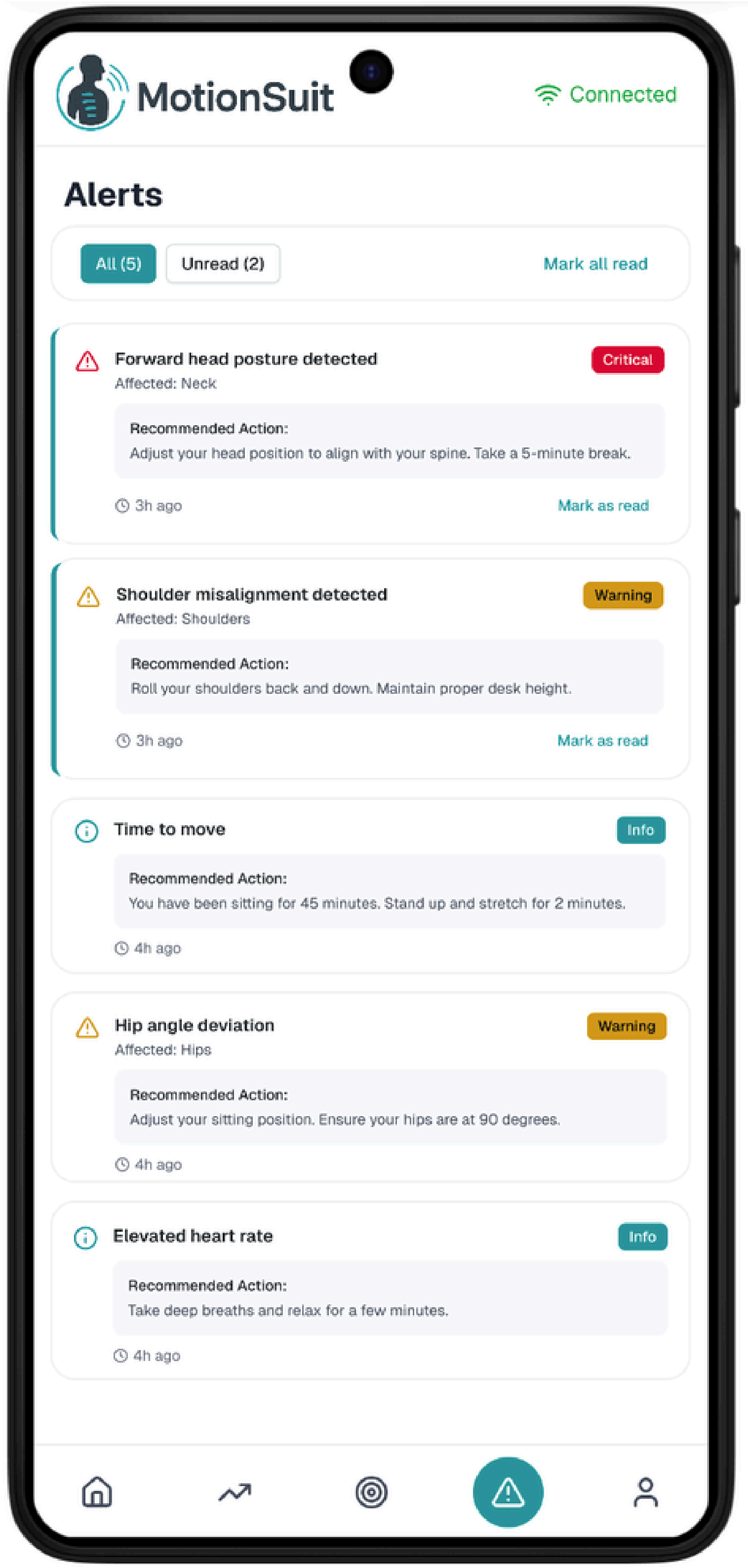
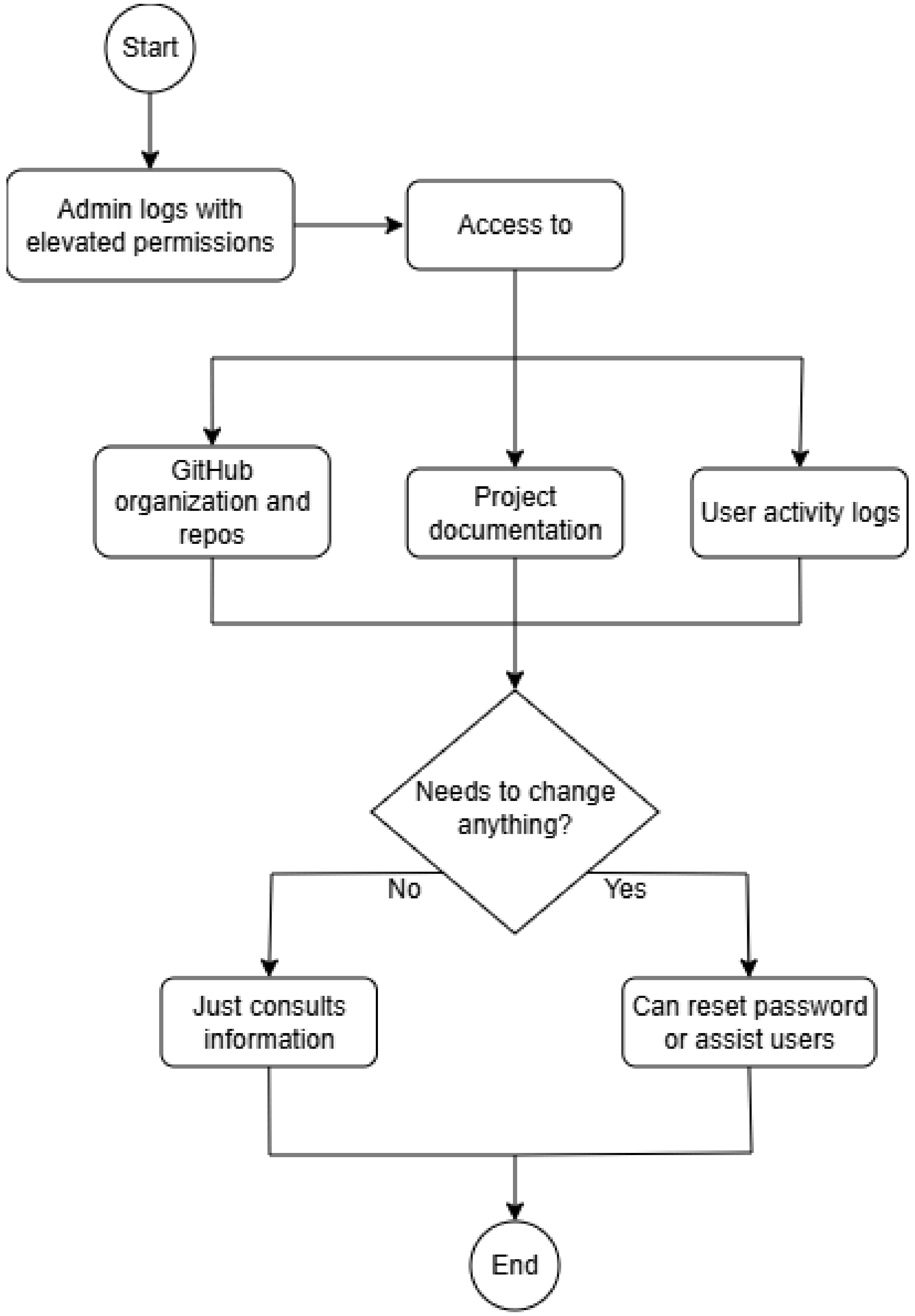
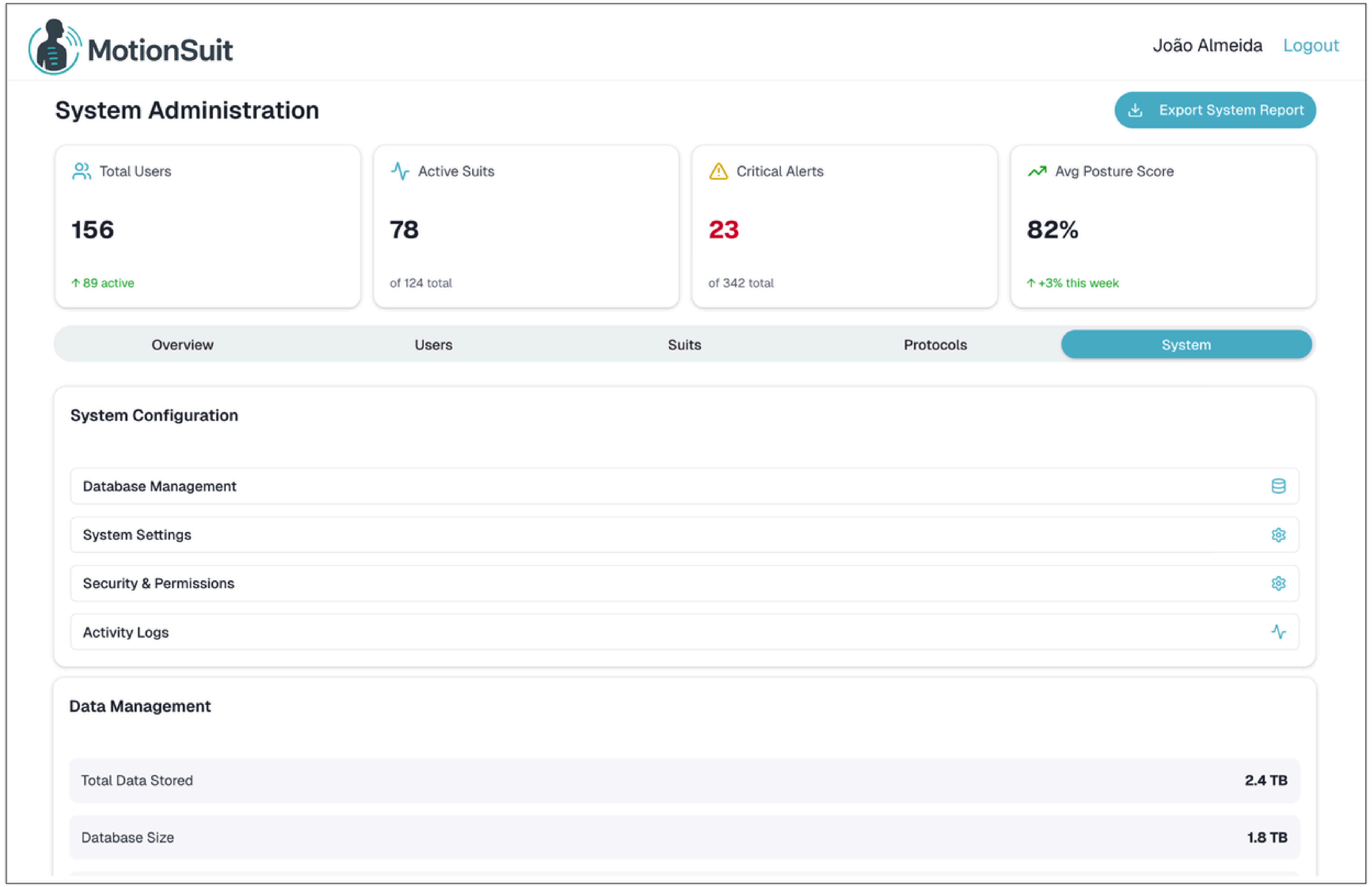






UC-9 → Access Full System Information





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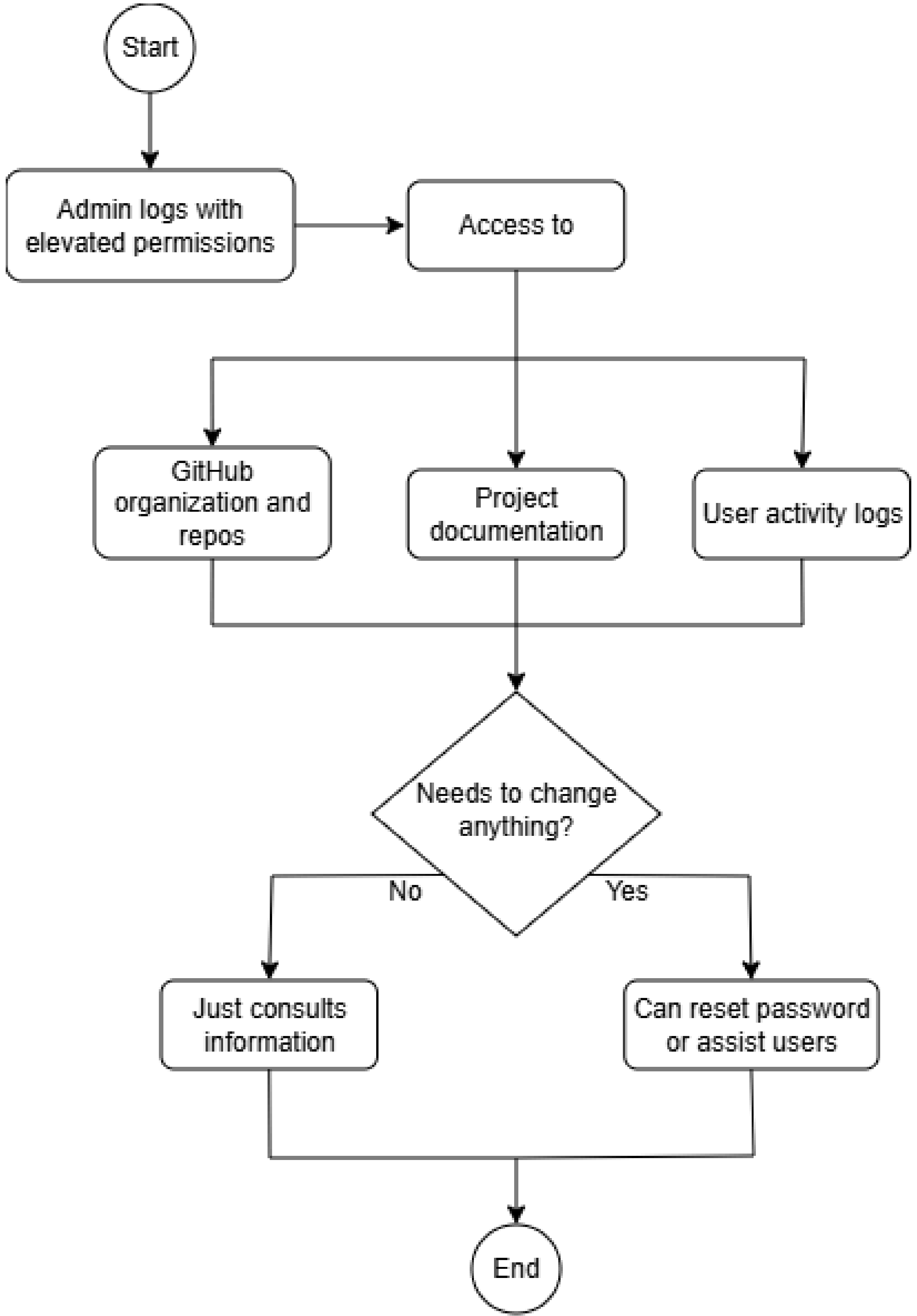

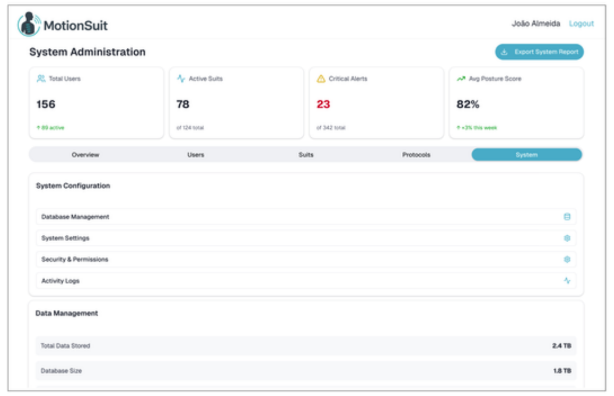

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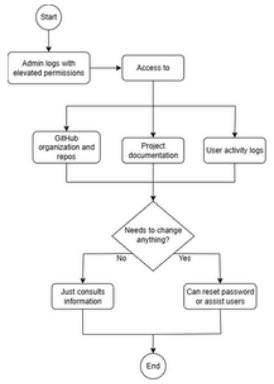
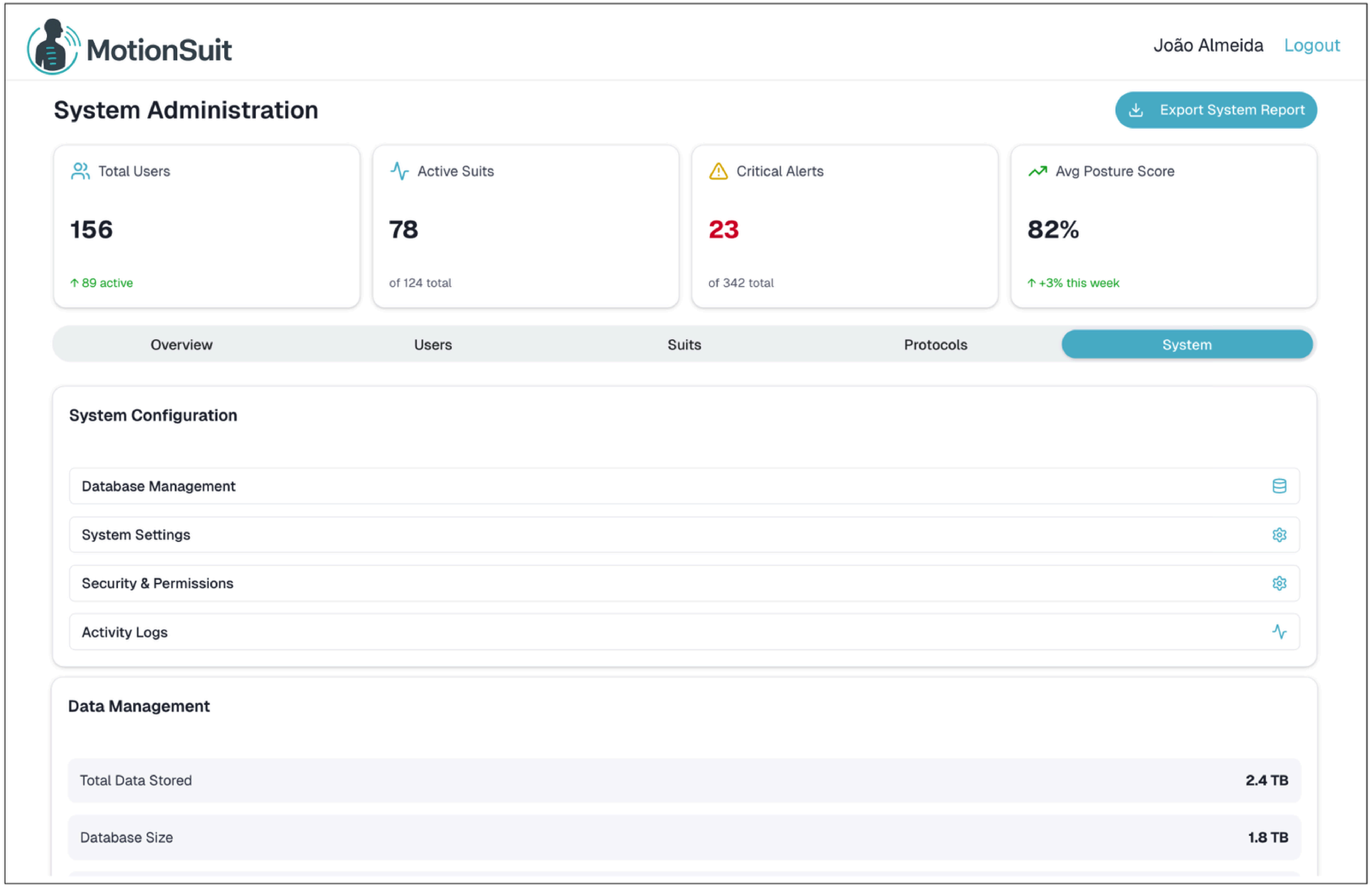
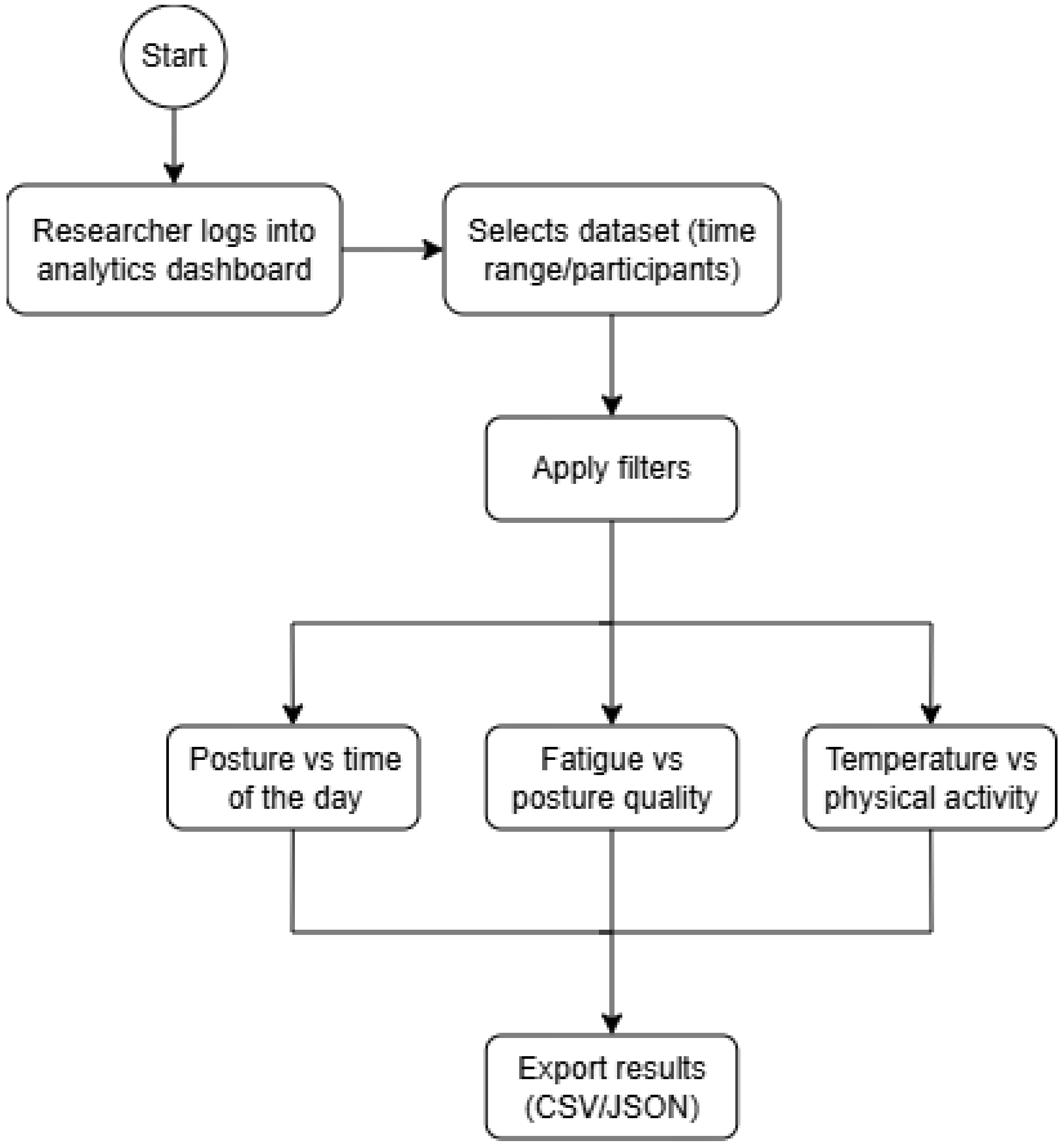
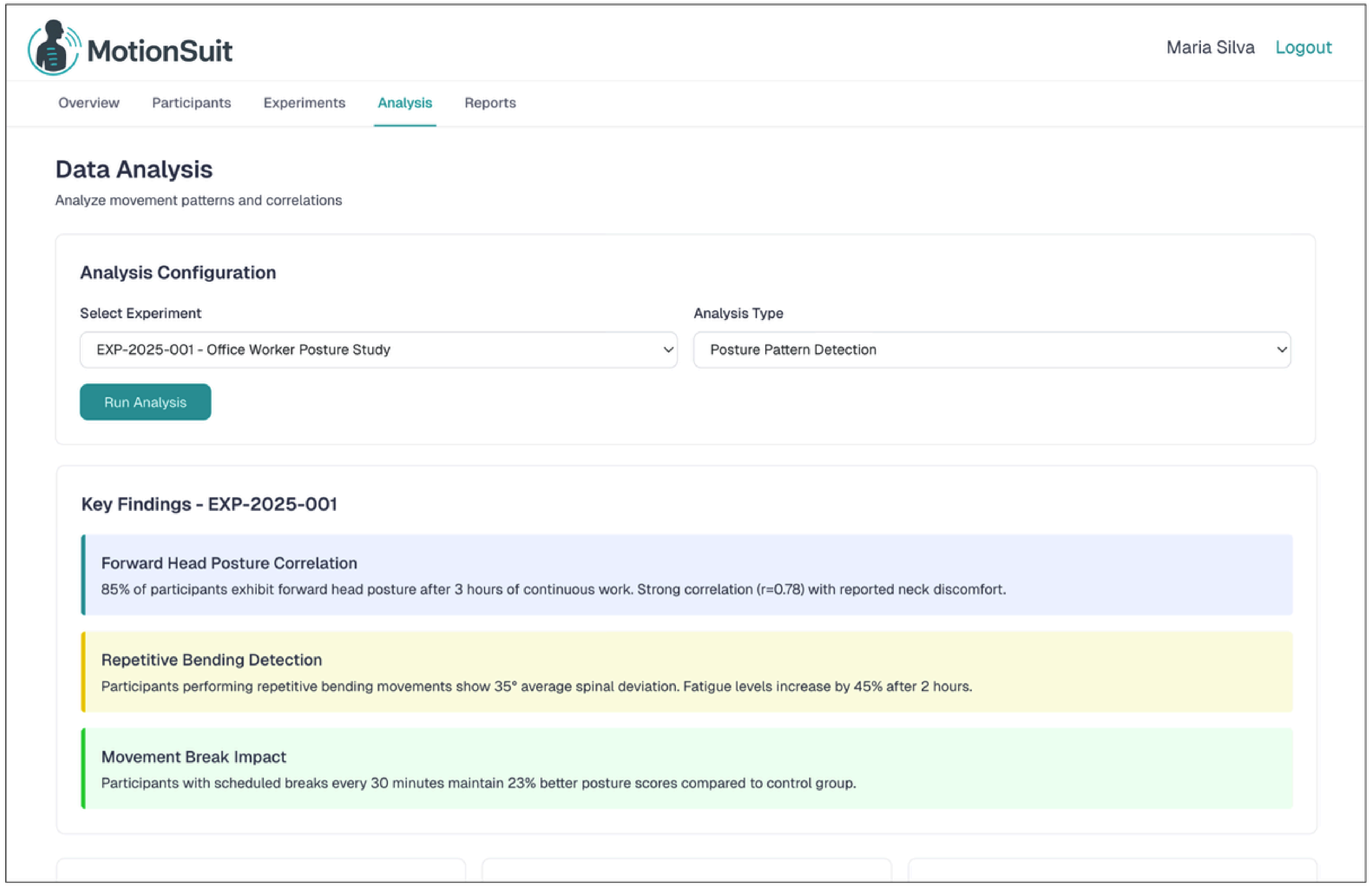




UC-11 → Analyse Captured Data





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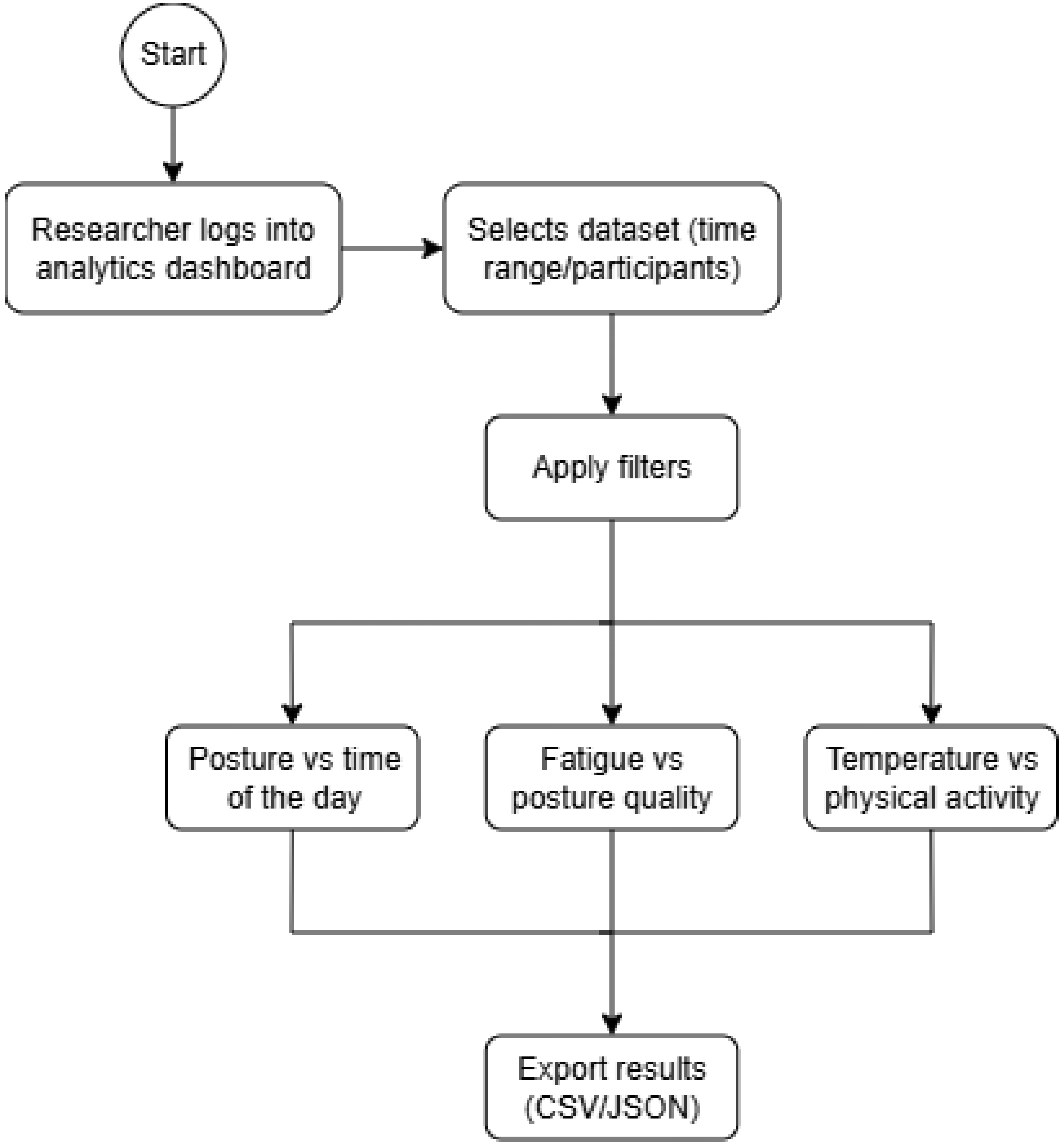

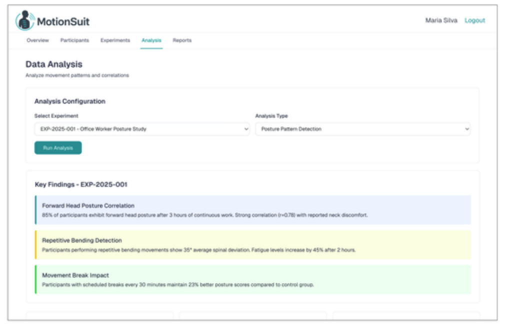

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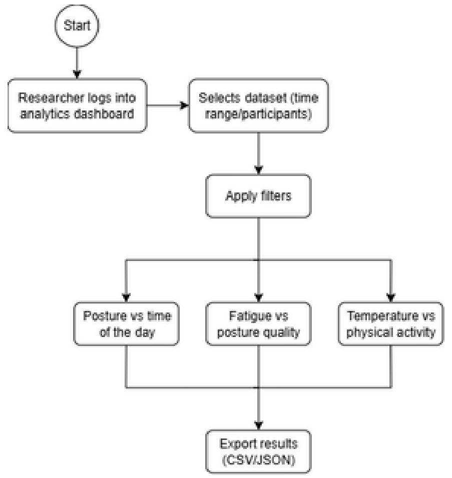

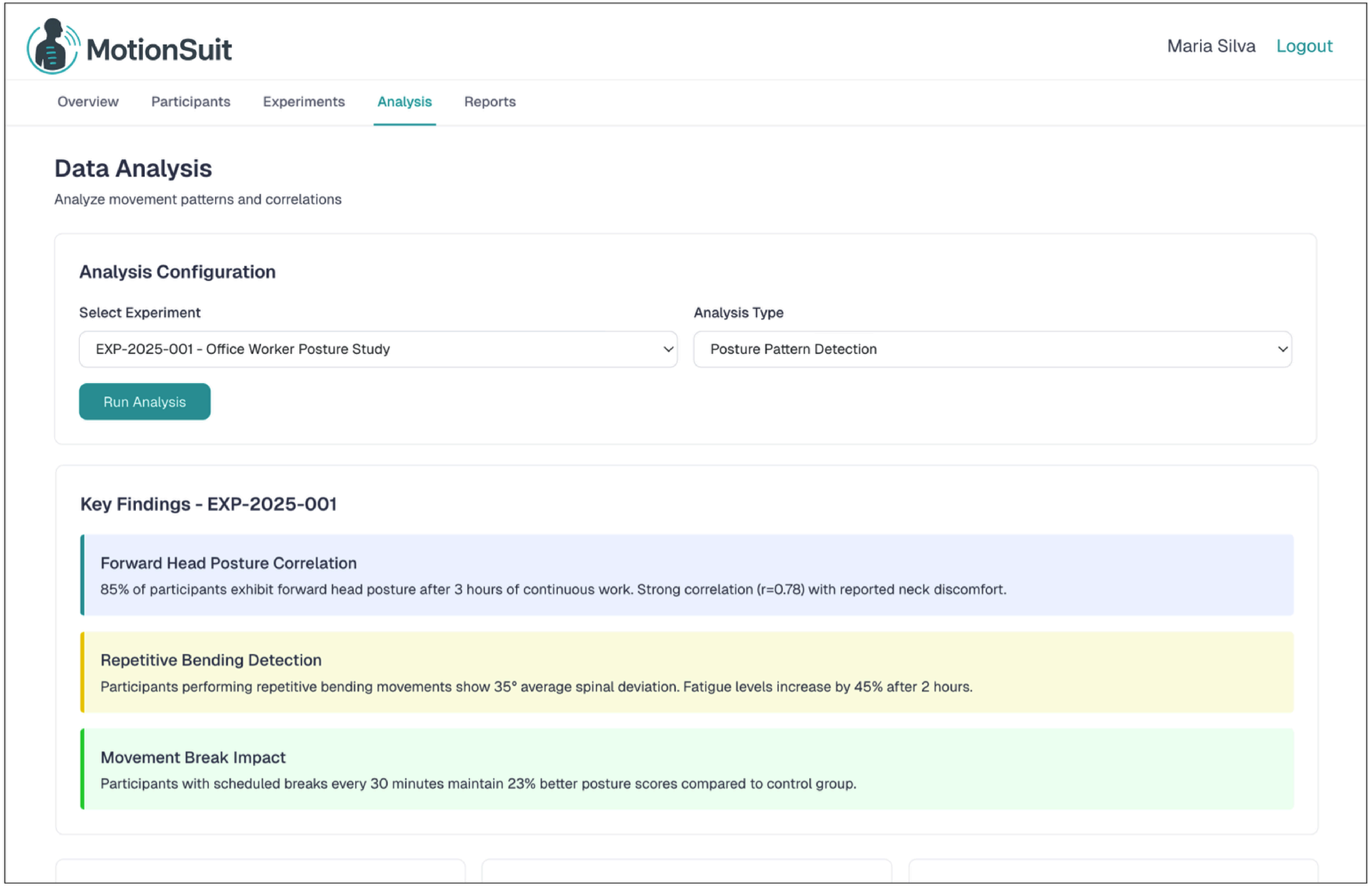

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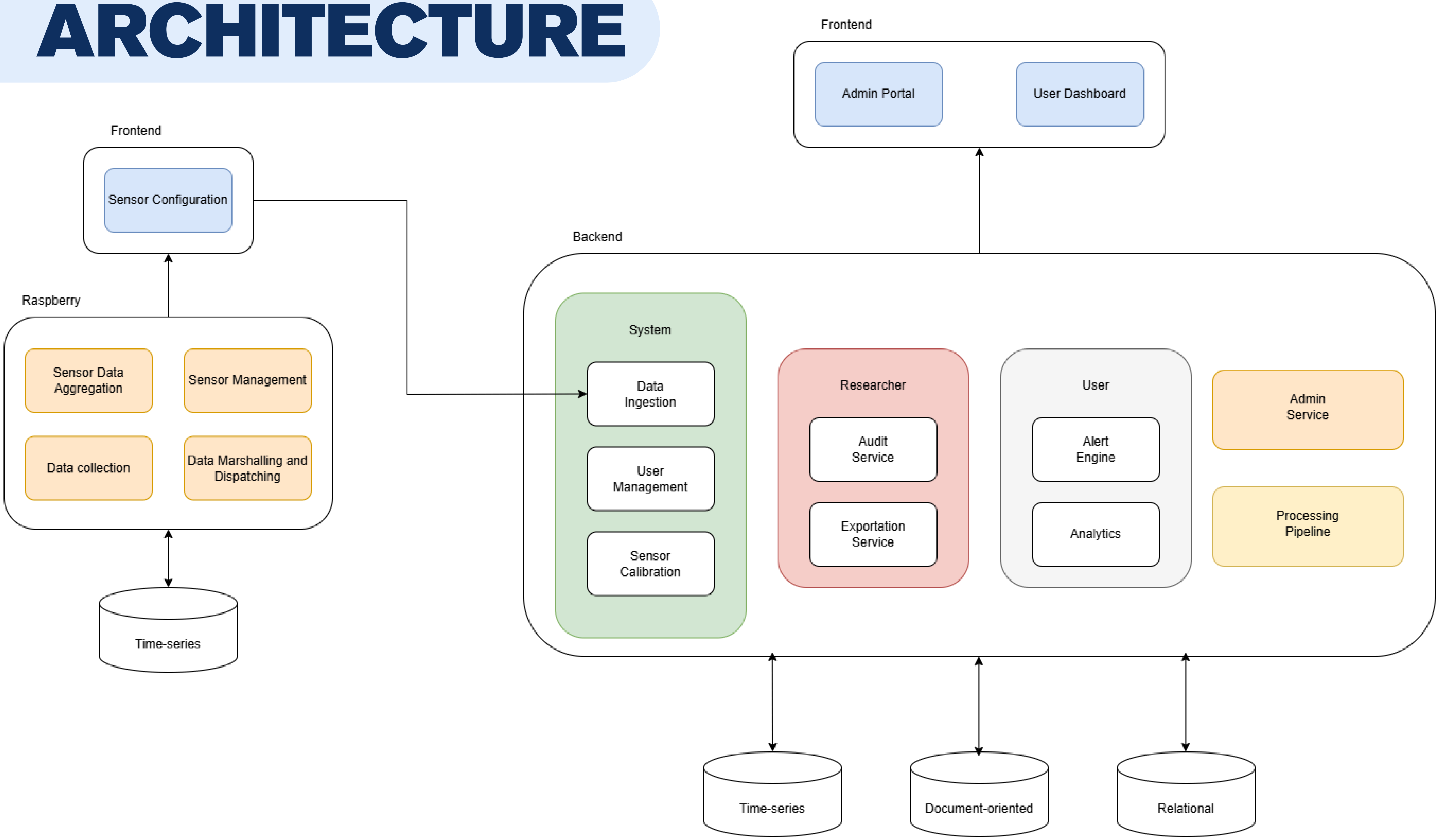

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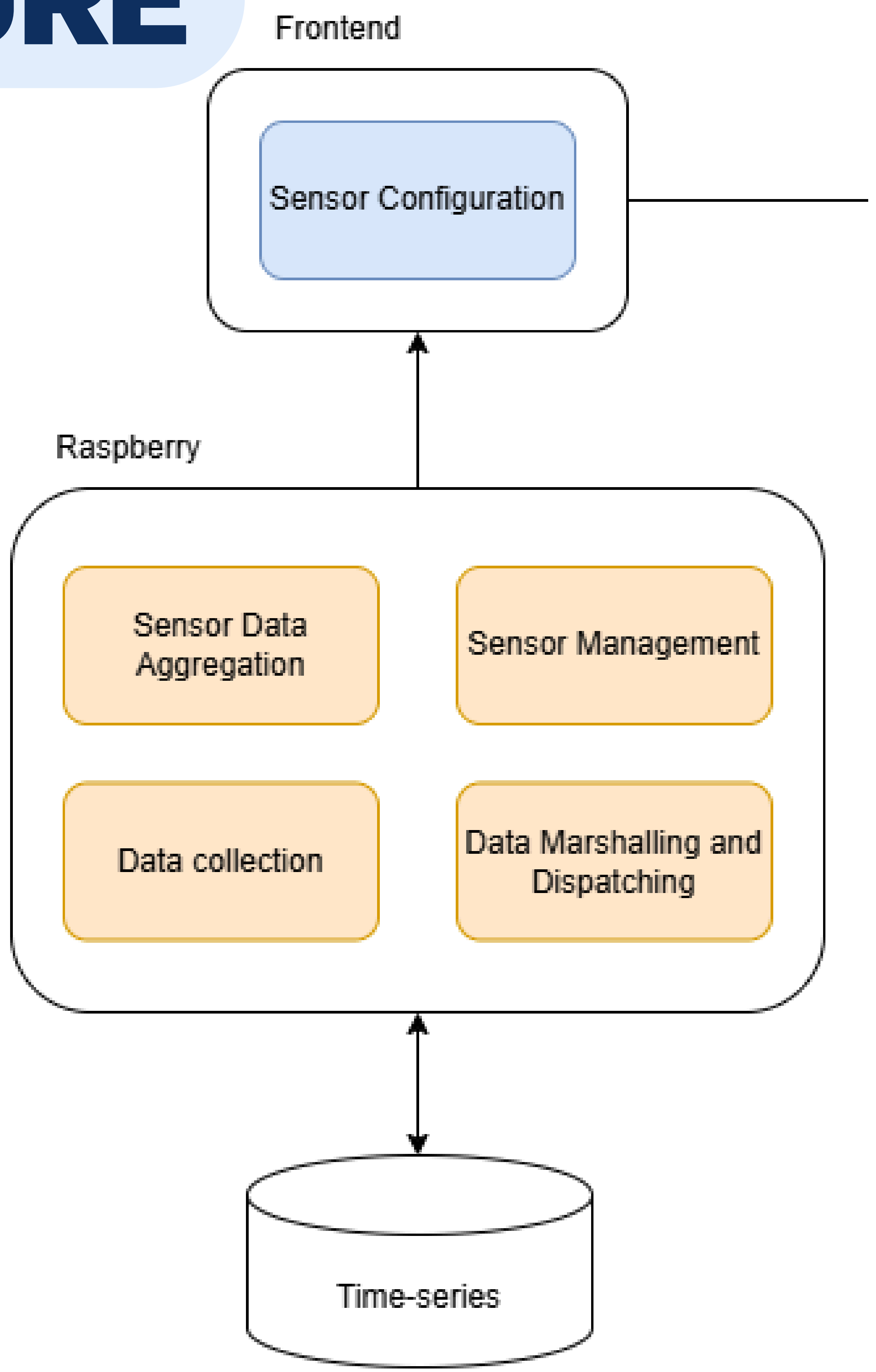
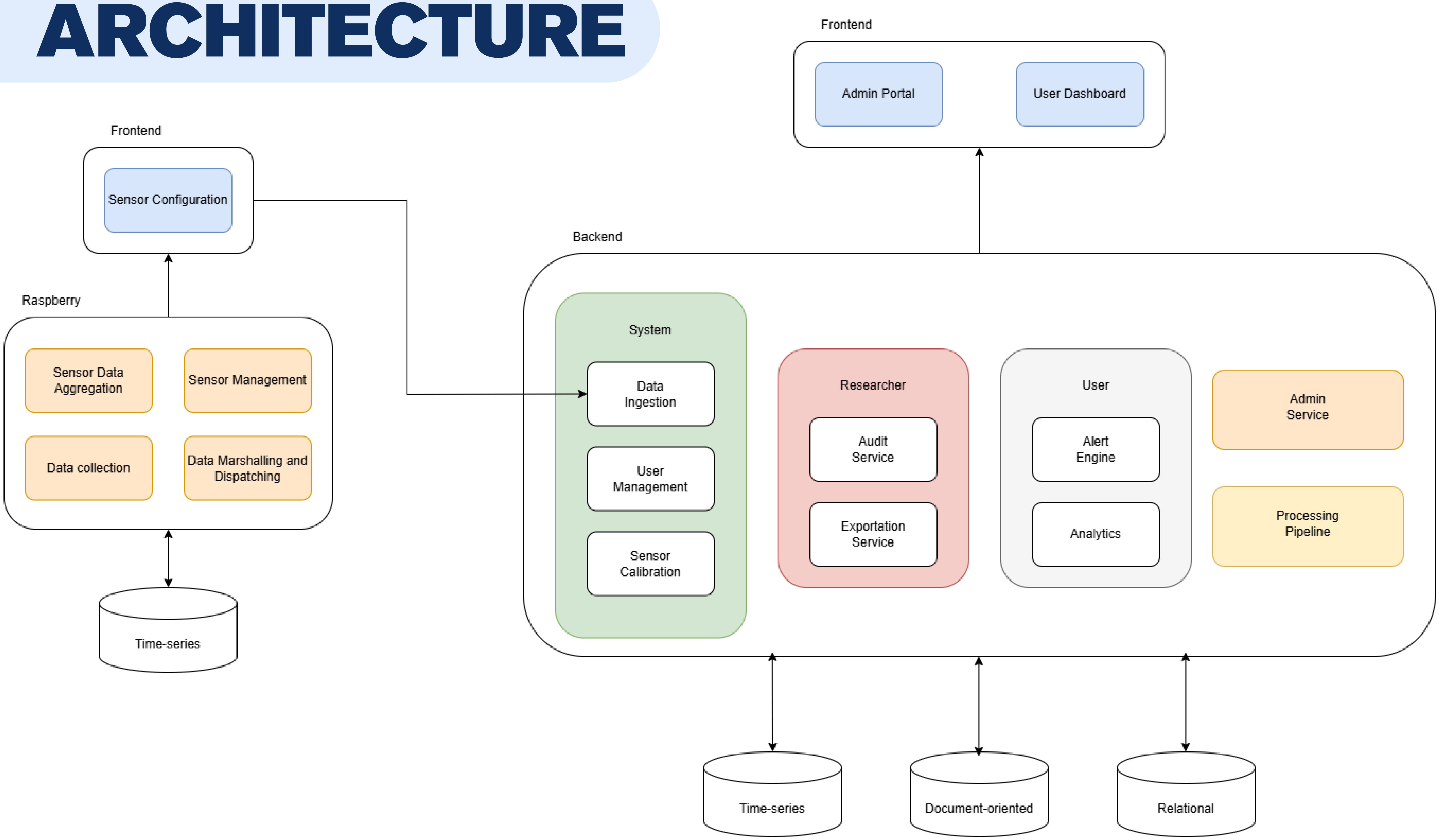


ARCHITECTURE

Sensor Data
SensorDataID
SessionID
Timestamp
Quaternion
Acceleration
Gyroscope
CalibrationSys
CalibrationGyro
CalibrationAccel
Euler
LinearAccel
Gravity
Temperature

ARCHITECTURE



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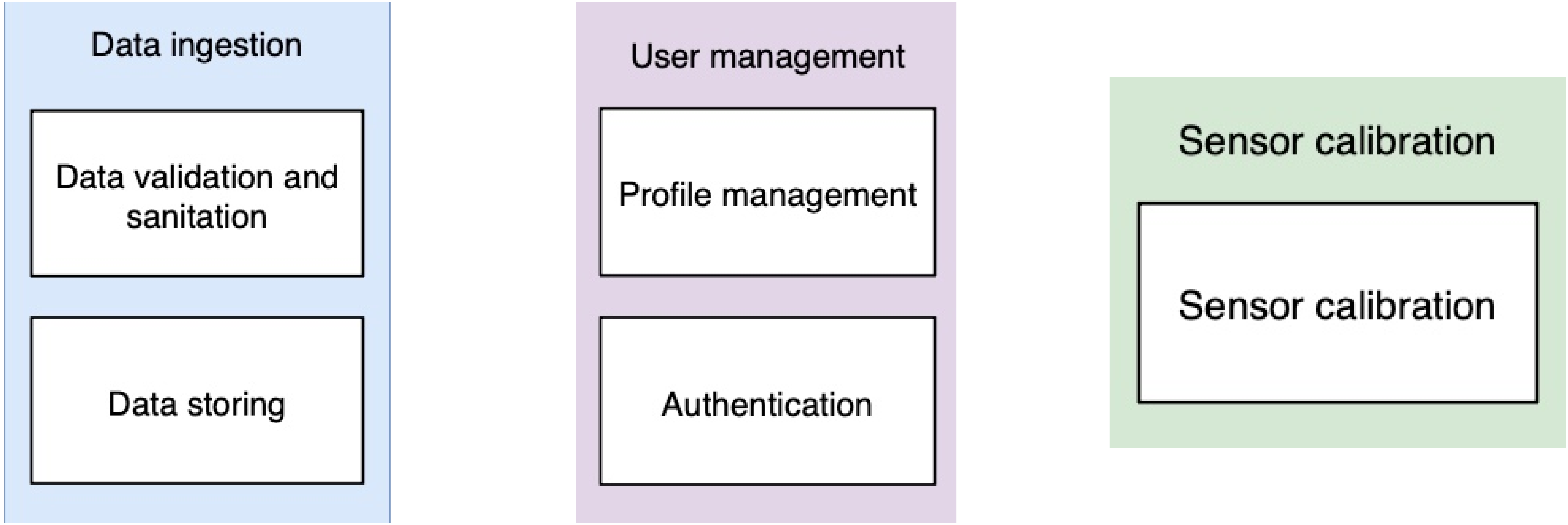

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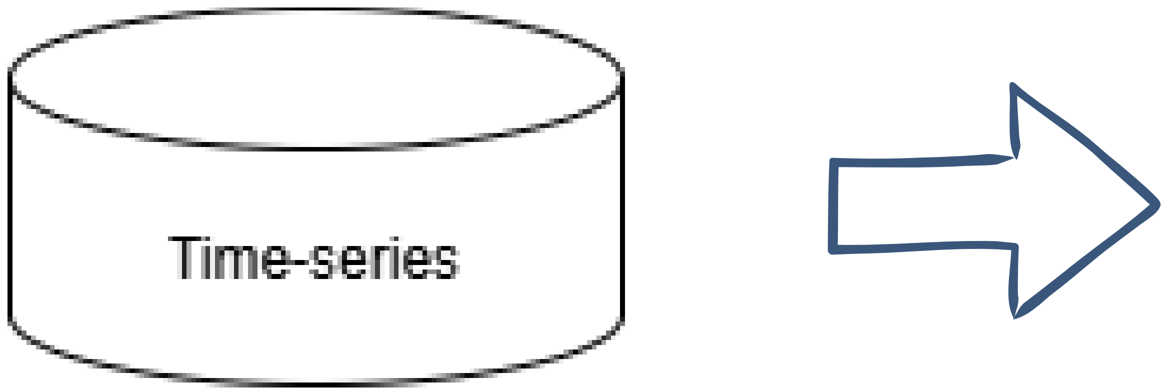
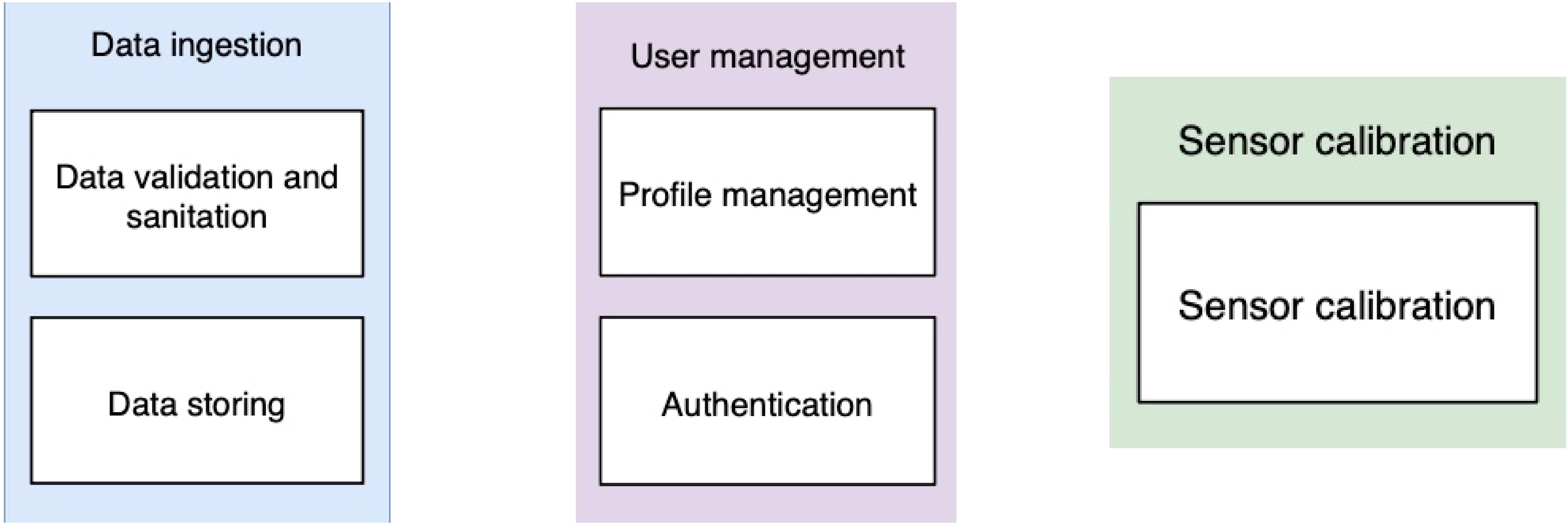


Sensor Data
SensorDataID
SessionID
Timestamp
Quaternion
Acceleration
Gyroscope
CalibrationSys
CalibrationGyro
CalibrationAccel
Euler
LinearAccel
Gravity
Temperature
IsValid

Session
SessionID
UserID
TimestampStart
TimestampEnd
AvgPostureQuality

Posture Alert
SessionID
AlertID
AlertType
Severity
Description
Timestamp

ARCHITECTURE



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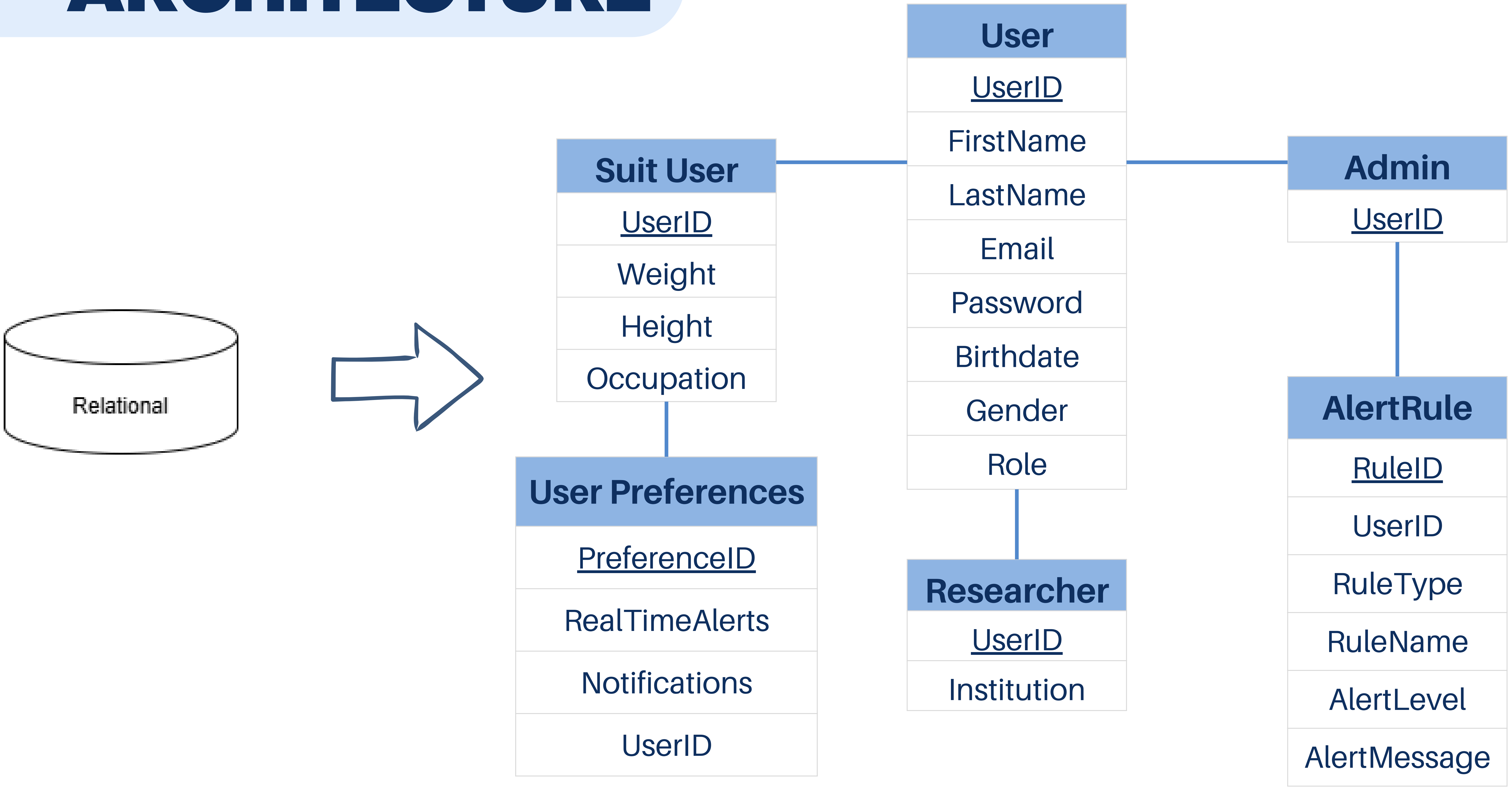

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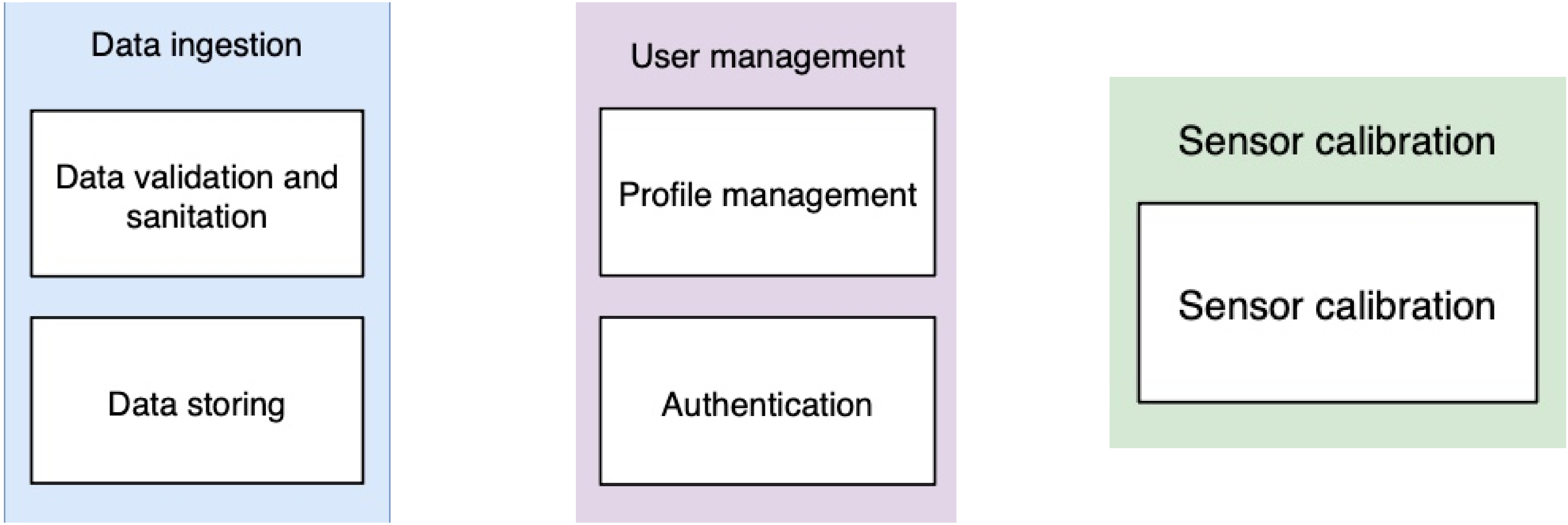

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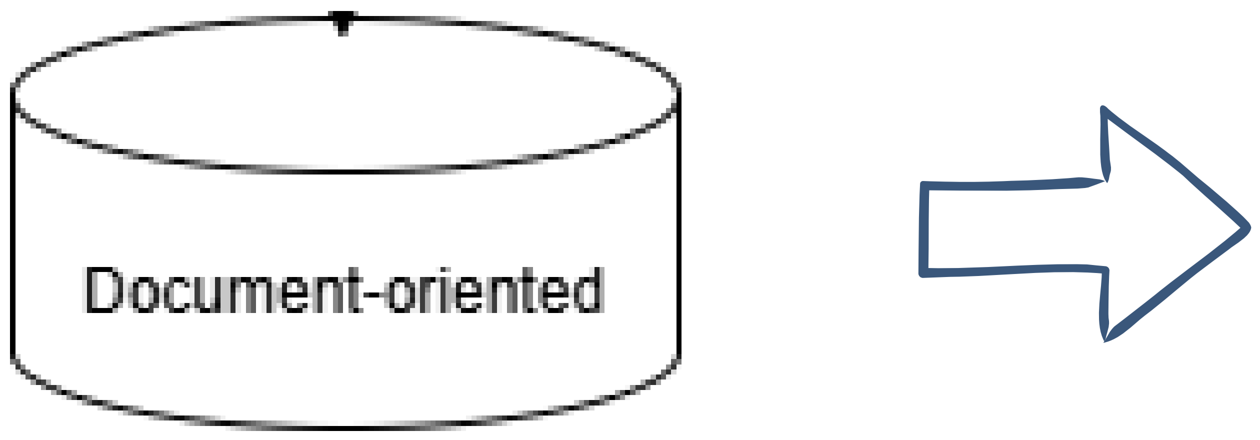
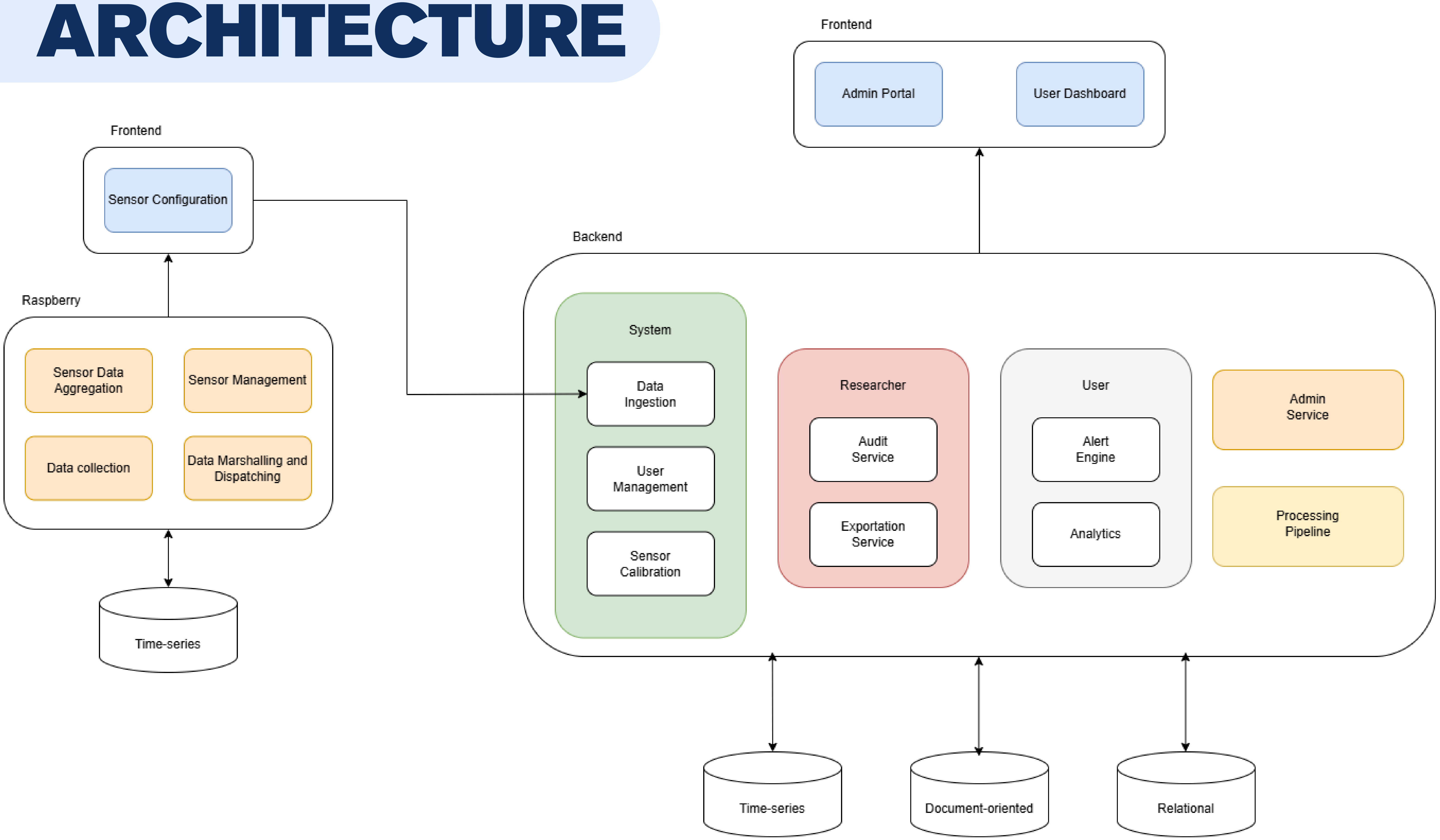


User Baseline
BaselineID
UserID
LastCalculatedAt
FatiguePercentage
IdealPosturePercentage
ShoulderAngleLimit
KneeAngleLimit
BackAngleLimit
NeckAngleLimit

ARCHITECTURE



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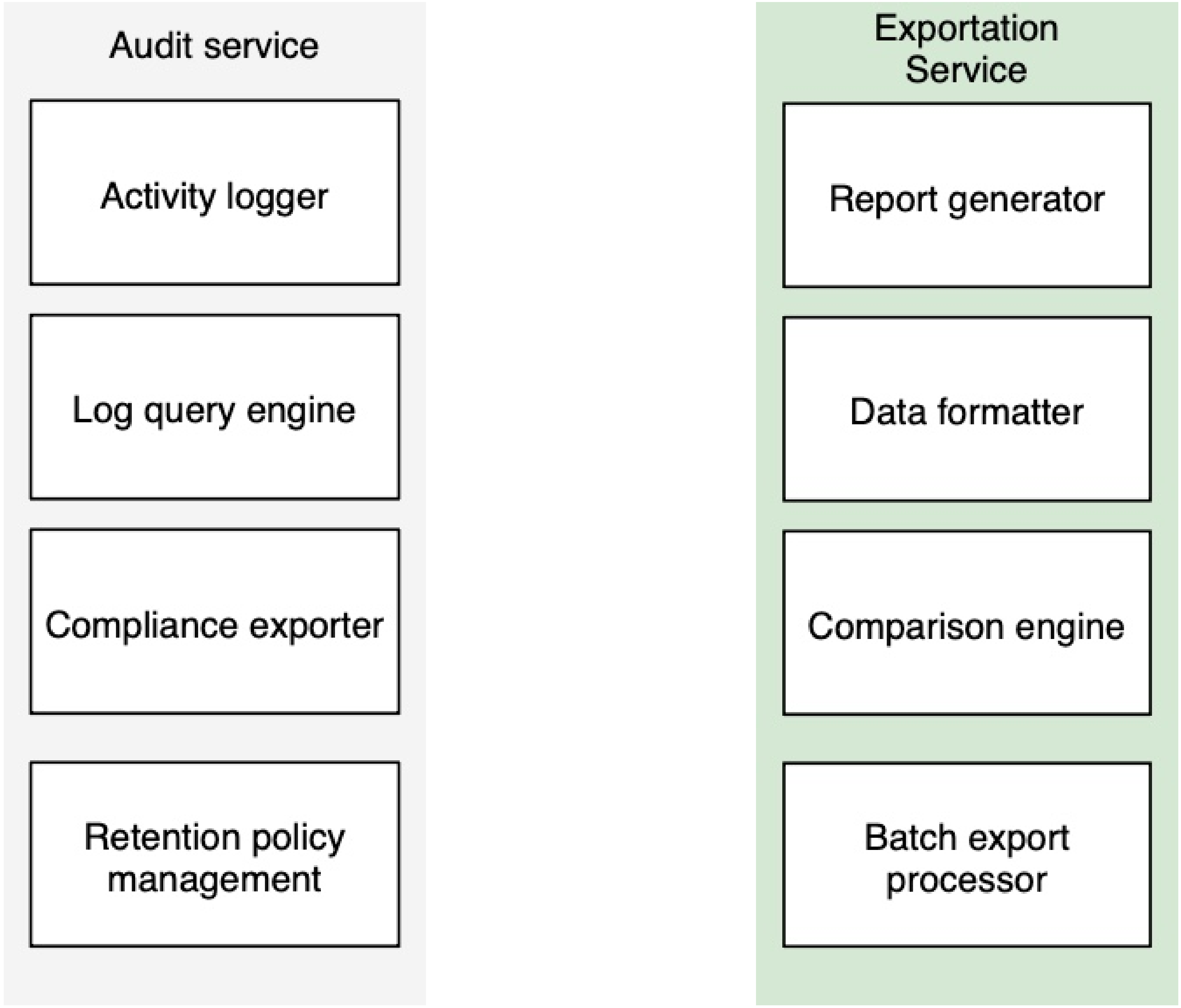

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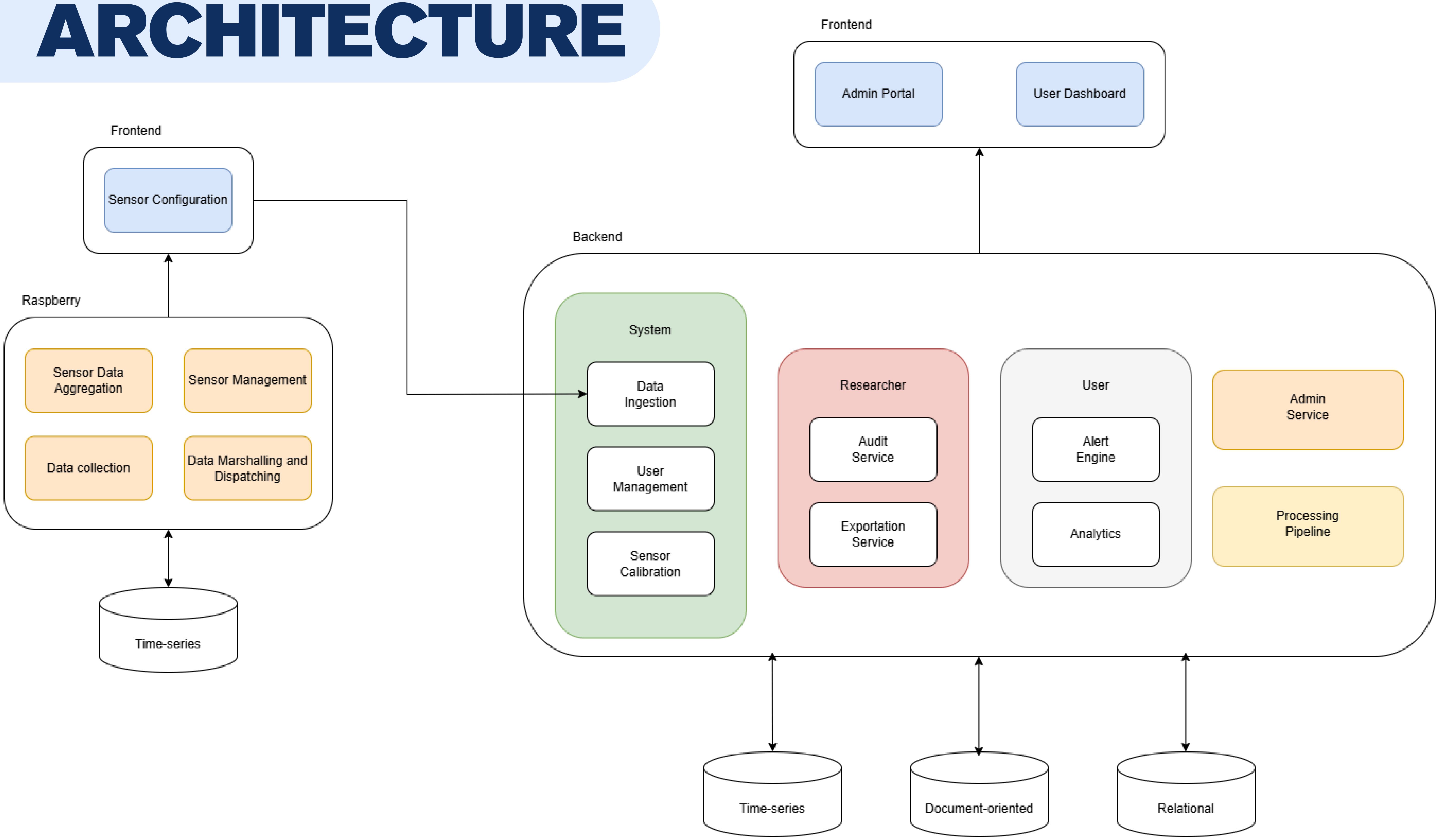

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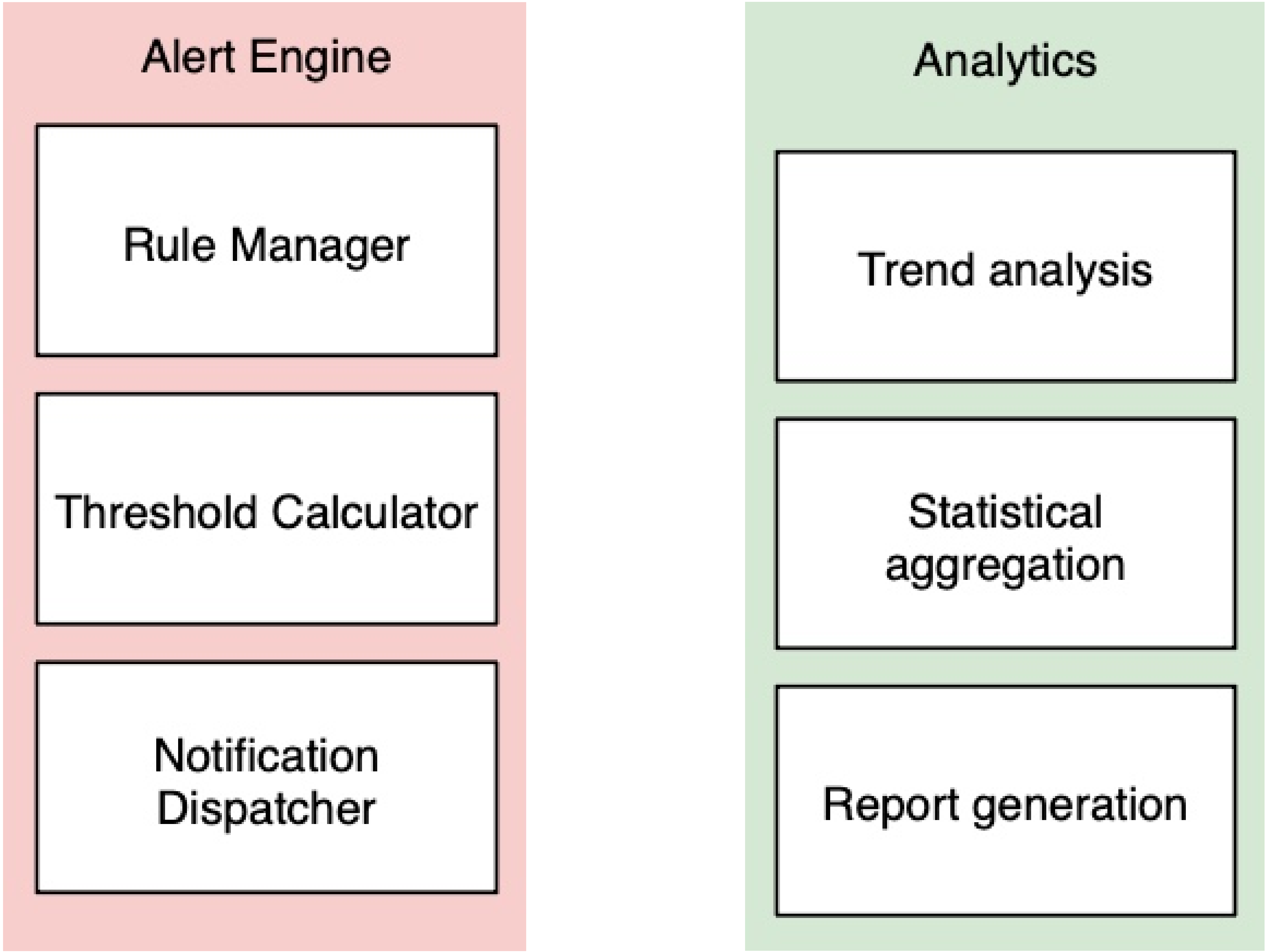

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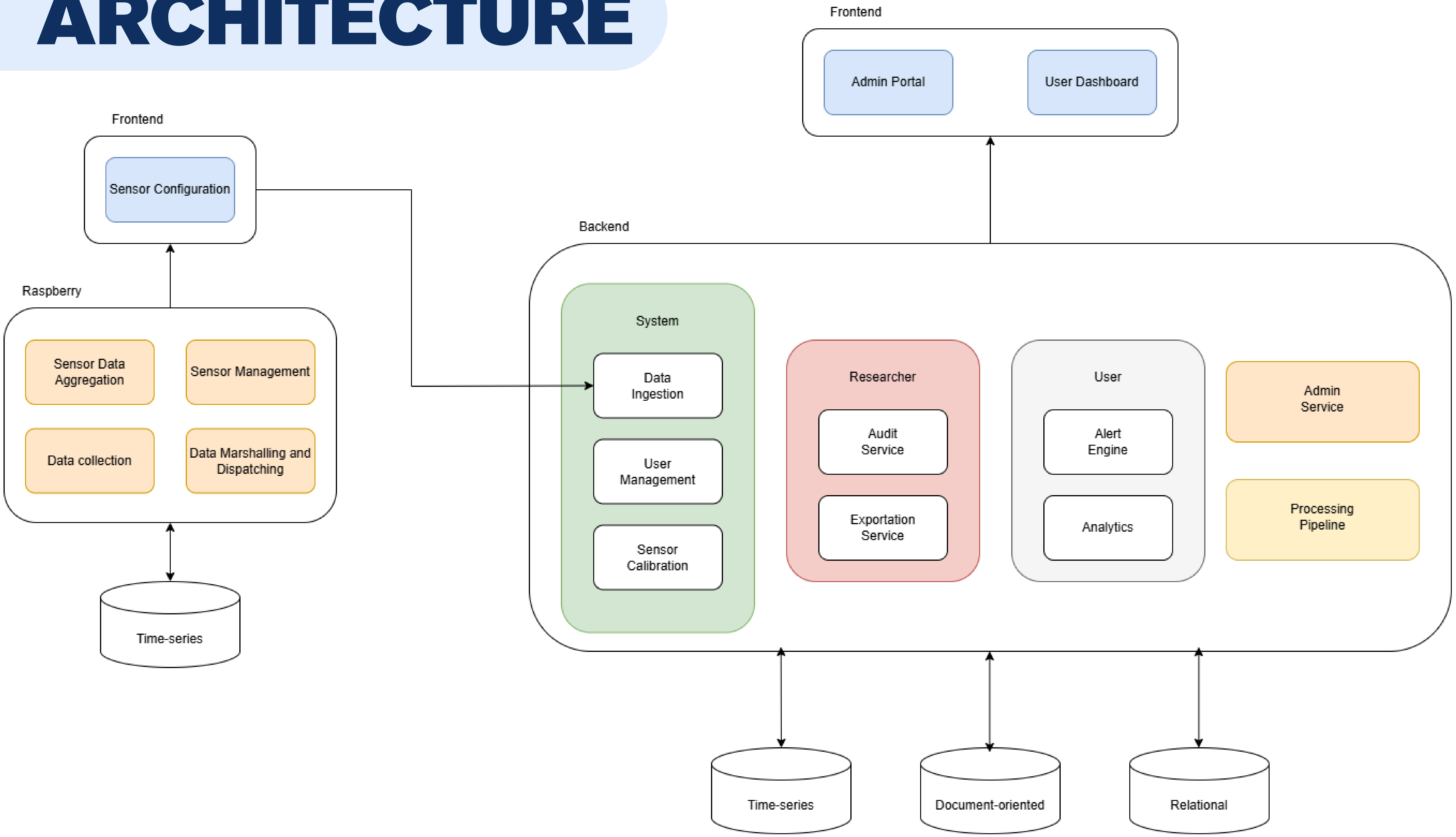

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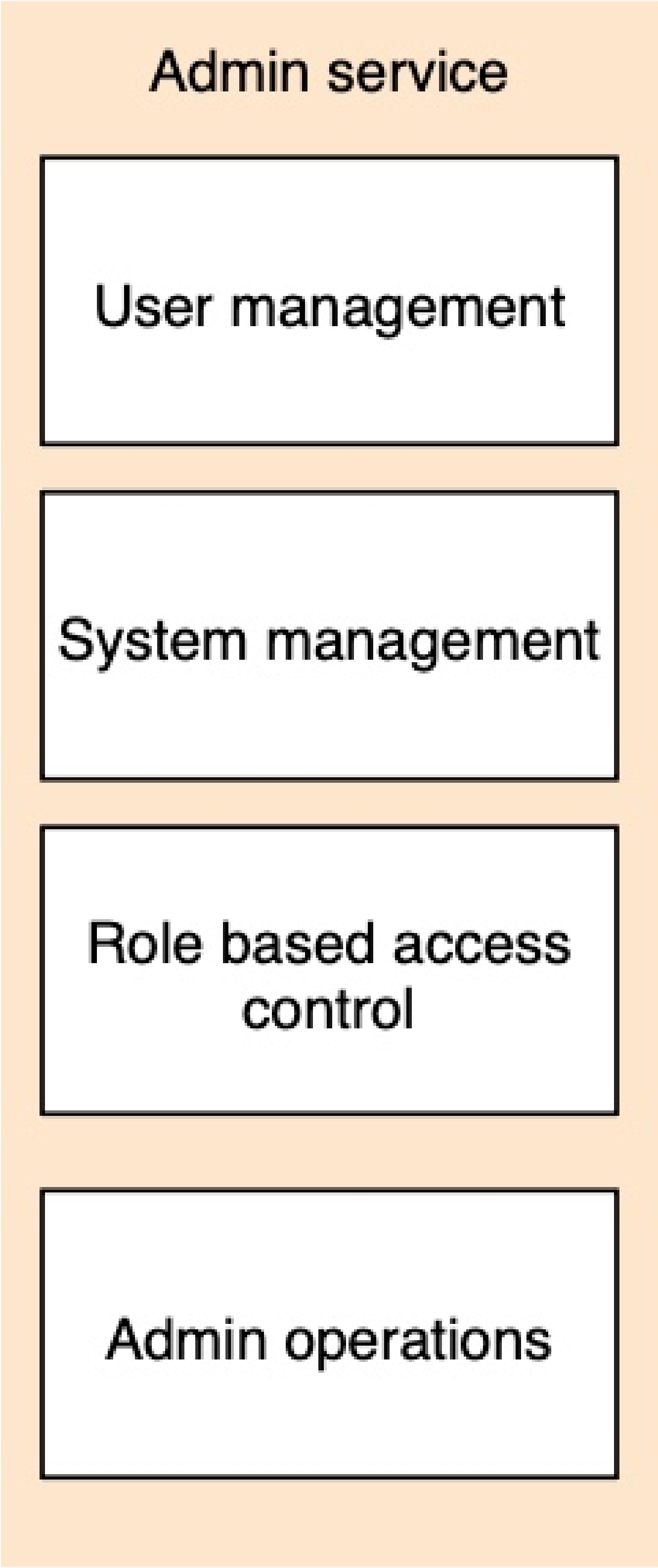

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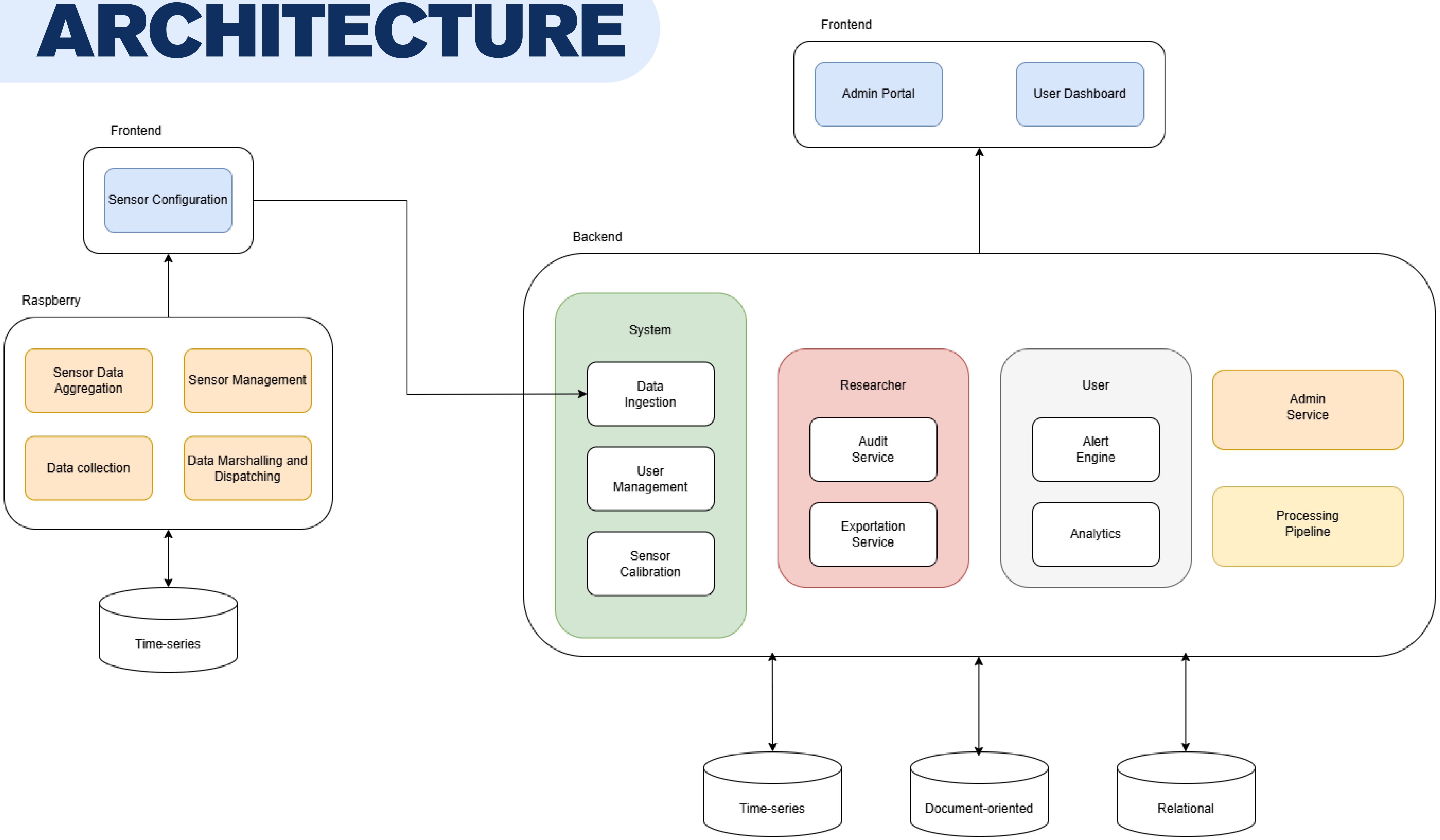

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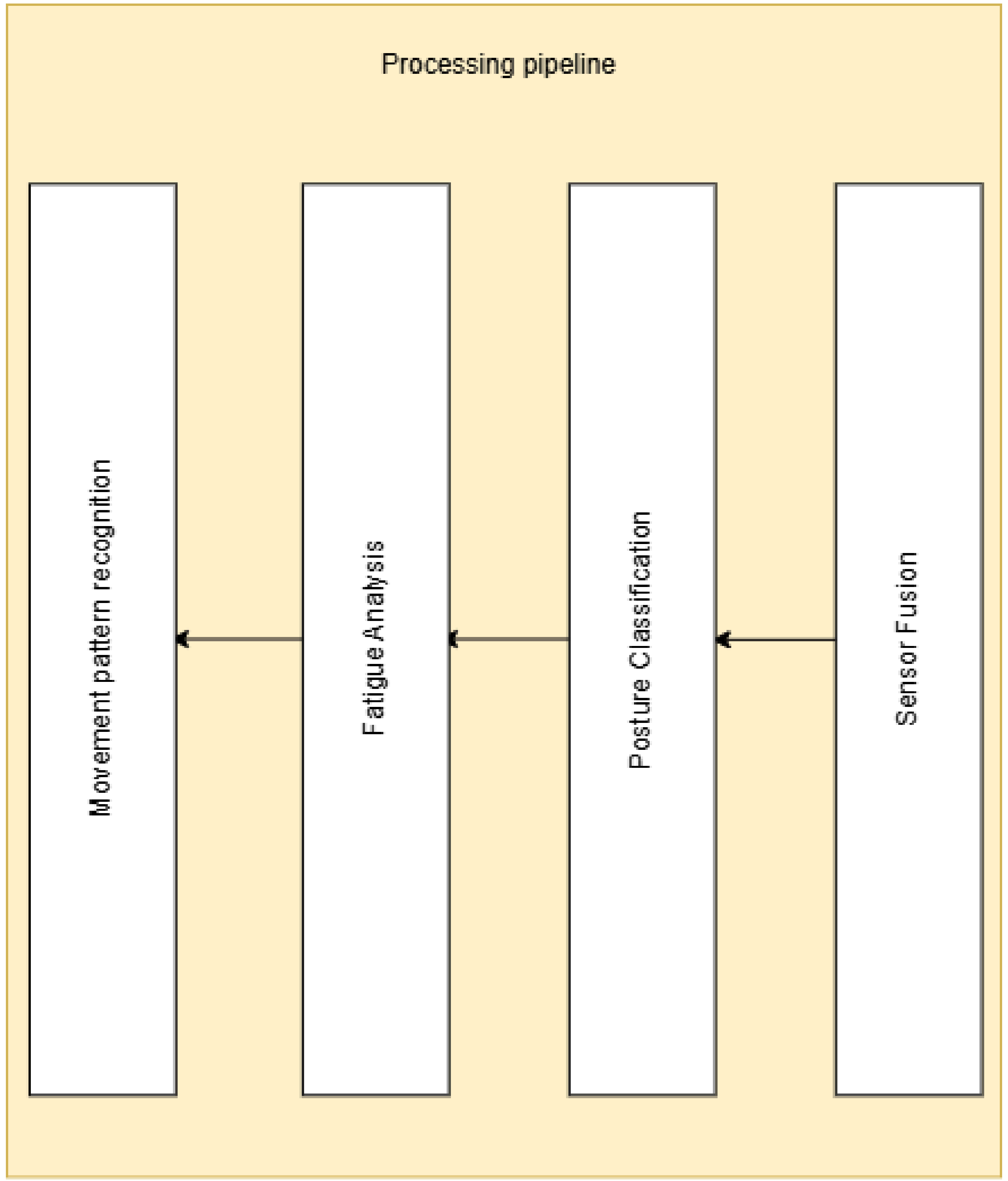

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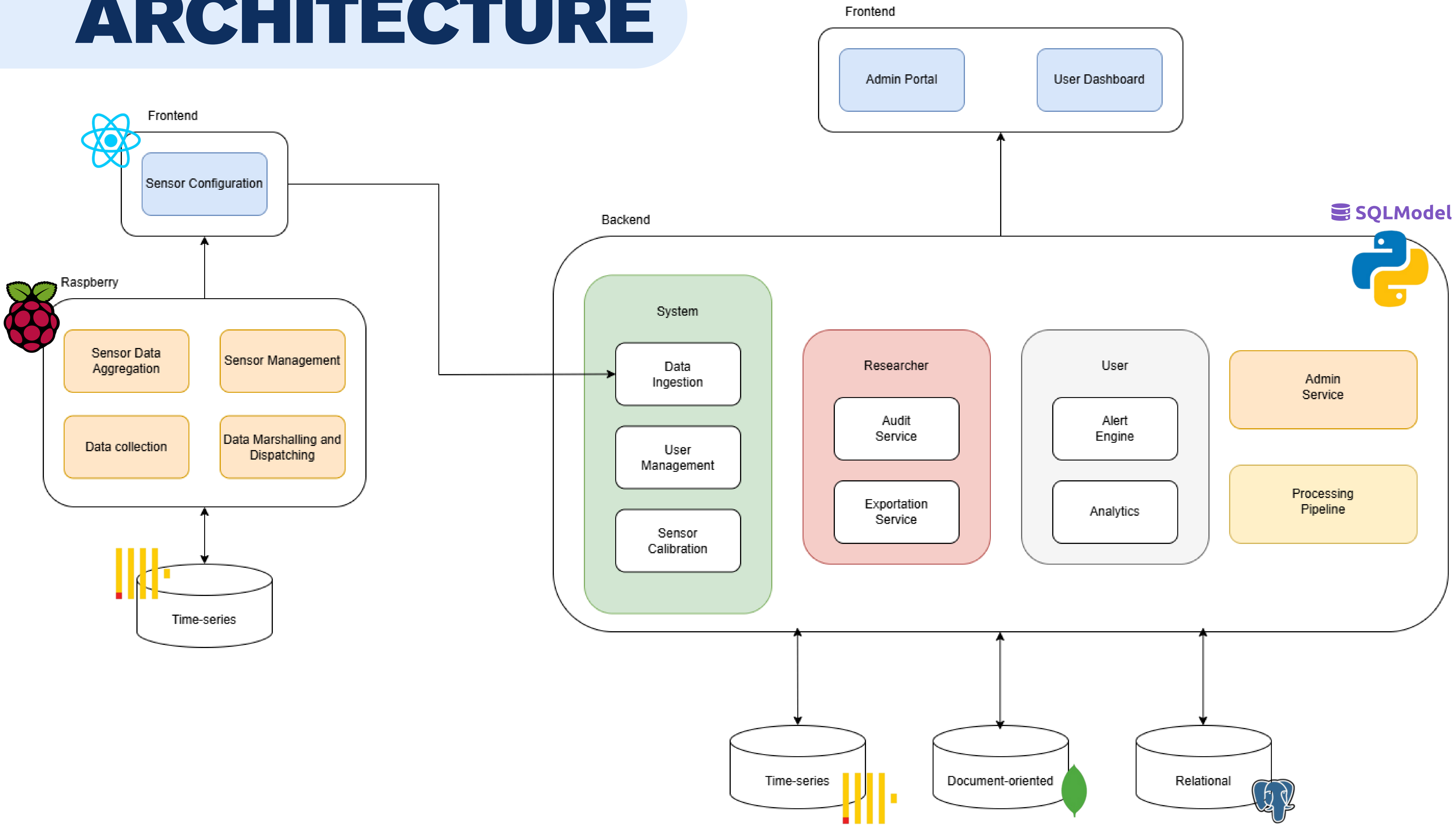
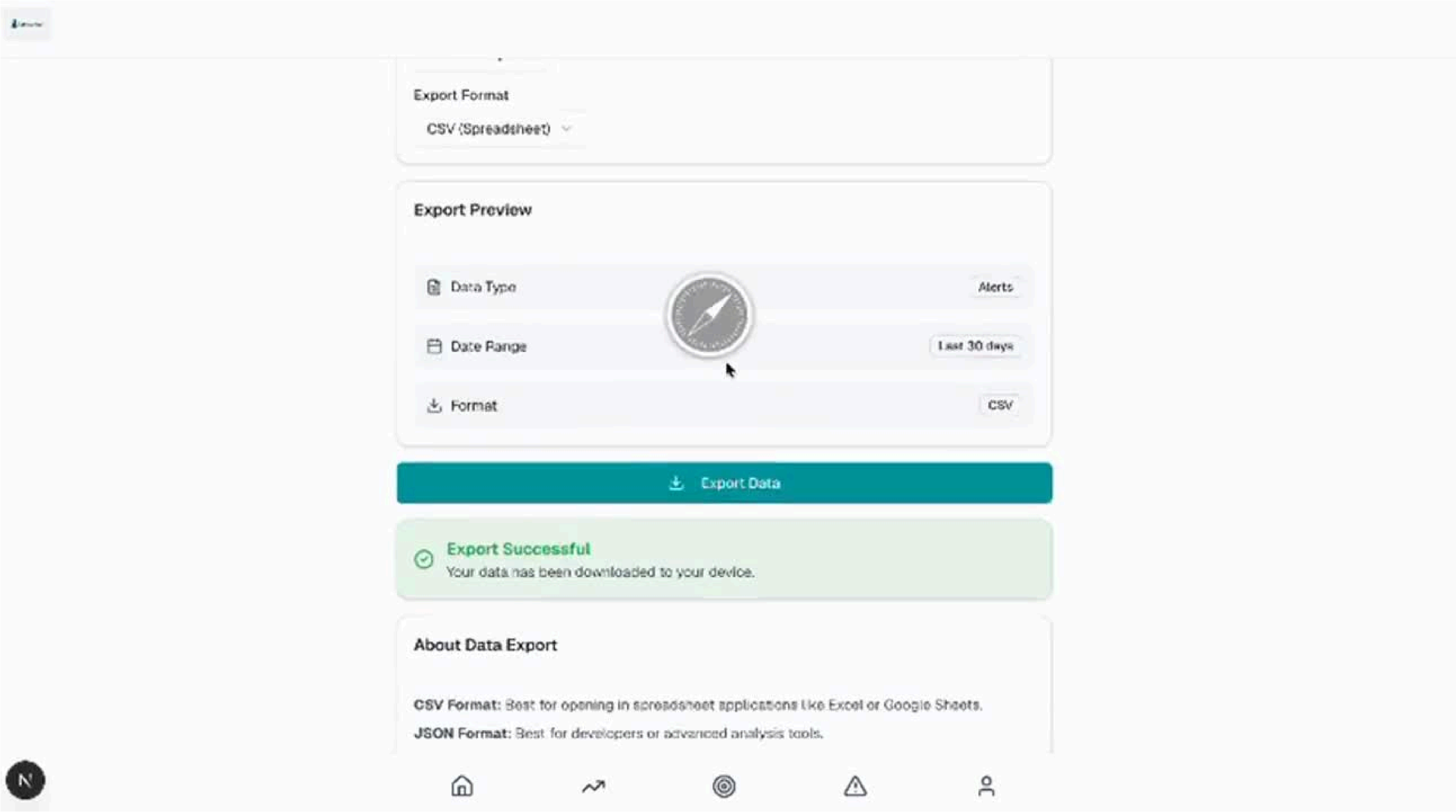
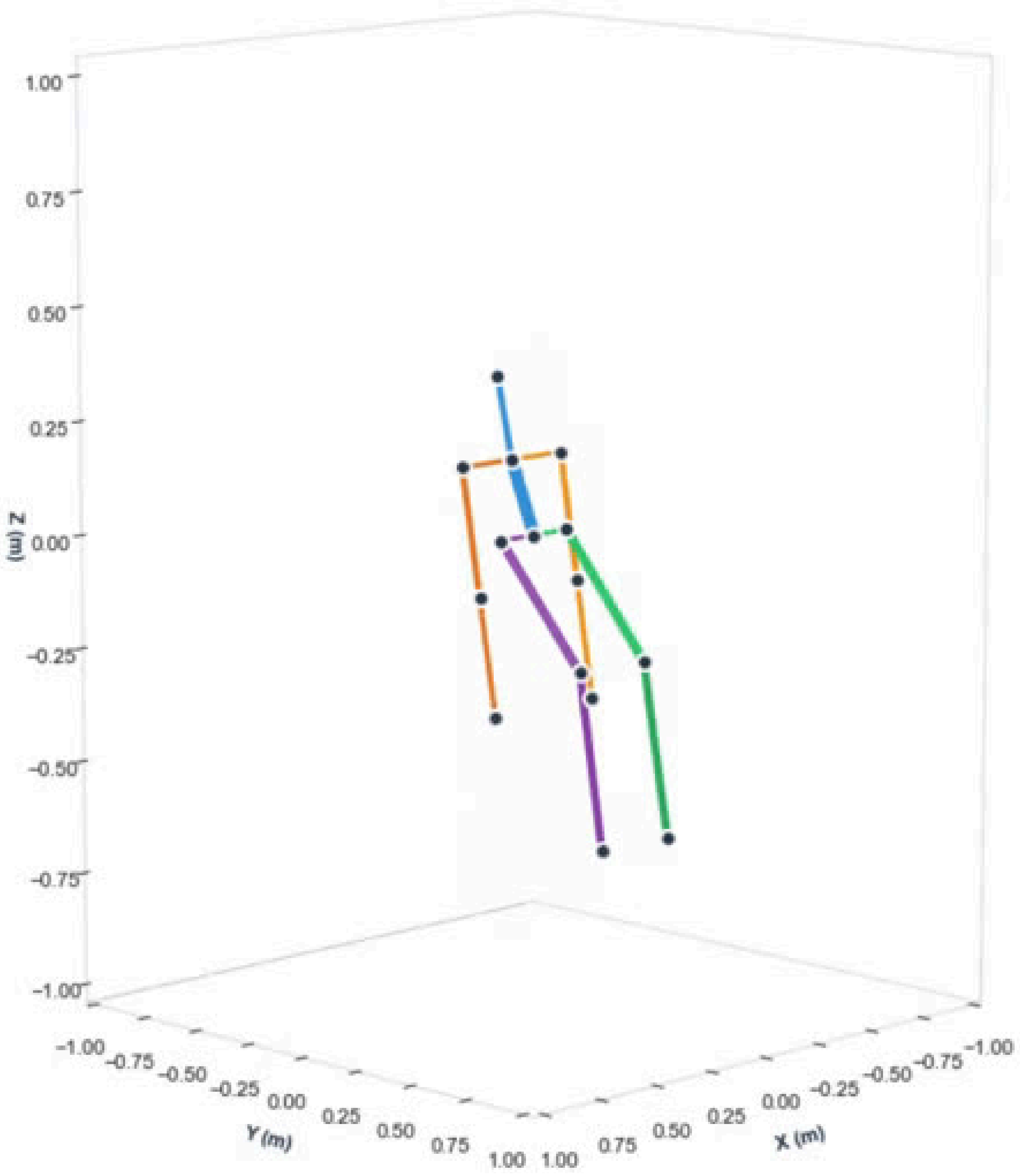


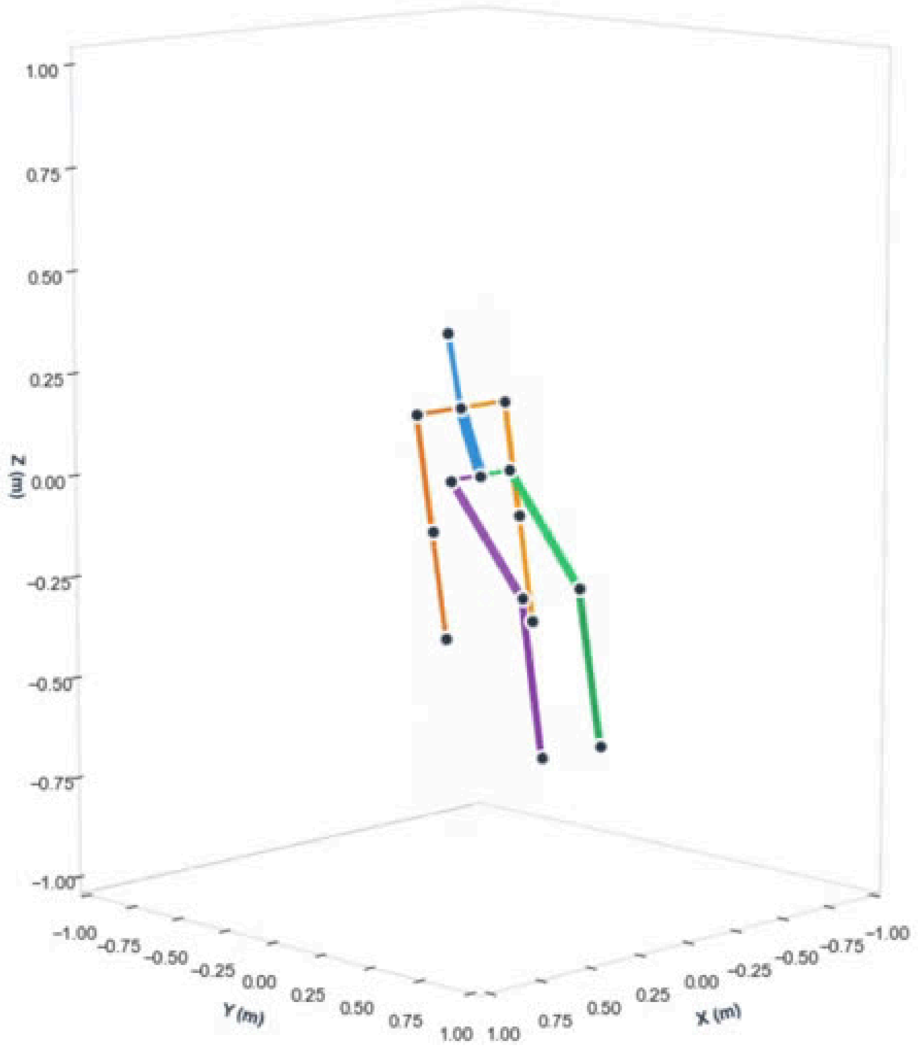
FRONTEND


Export Format
CSV (Spreadsheet)
Export Preview
Data Type
Alerts
Date Range
Last 30 days
Format
CSV
Export Data
Export Successful
Your data has been downloaded to your device.
About Data Export
CSV Format: Best for opening in spreadsheet applications like Excel or Google Sheets.
JSON Format: Best for developers or advanced analysis tools.

SENSORS



Z (m)
1.00
0.75
0.50
0.25
0.00
-0.25
-0.50
-0.75
-1.00
-1.00 -0.75 -0.50 -0.25 0.00 0.25 0.50 0.75 1.00
Y (m)
1.00 0.75 0.50 0.25 0.00 -0.25 -0.50 -0.75 -1.00
X (m)



THANK YOU!

https://pei-motionsuit.github.io/microsite/